Lecture 18: Bonding in transition metal compounds and coordination complexes-3

Jin Xie (谢琎), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/12/24

Major points of last lecture

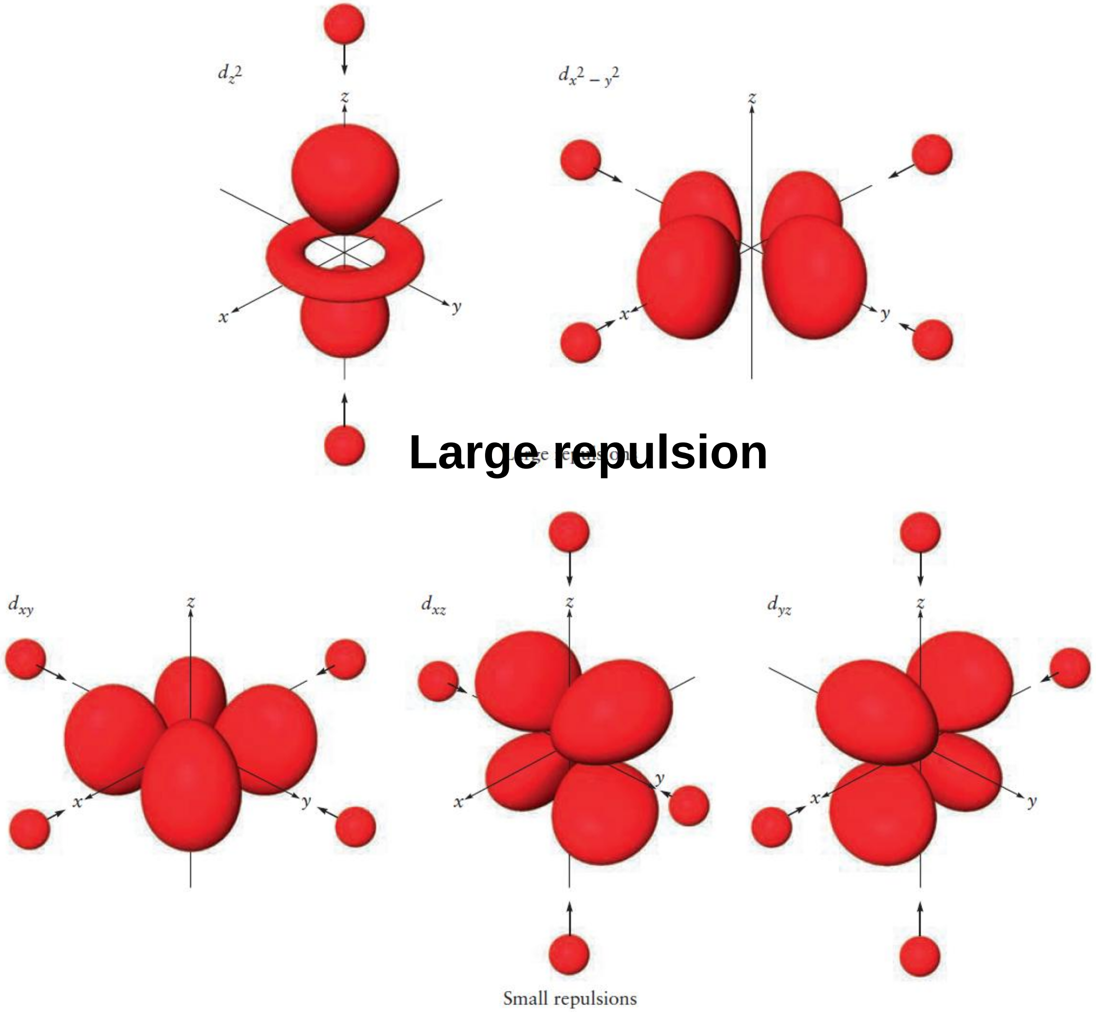
- **Bonding in transition metal compounds and coordination complexes**
  1. **Chemistry of the Transition Metals**
  2. **Introduction to Coordination Chemistry**
  3. **Structures of Coordination Complexes**
  4. **Crystal Field Theory: Optical and Magnetic Properties**
  5. **Optical Properties and the Spectrochemical Series**
  6. **Bonding in Coordination Complexes**

Crystal Field Theory: Optical and Magnetic Properties

Crystal field theory (晶体场理论): an **ionic description** of **metal–ligand bonding**.

A simple/useful model for understanding the electronic structure, optical properties, and magnetic properties of coordination complexes.

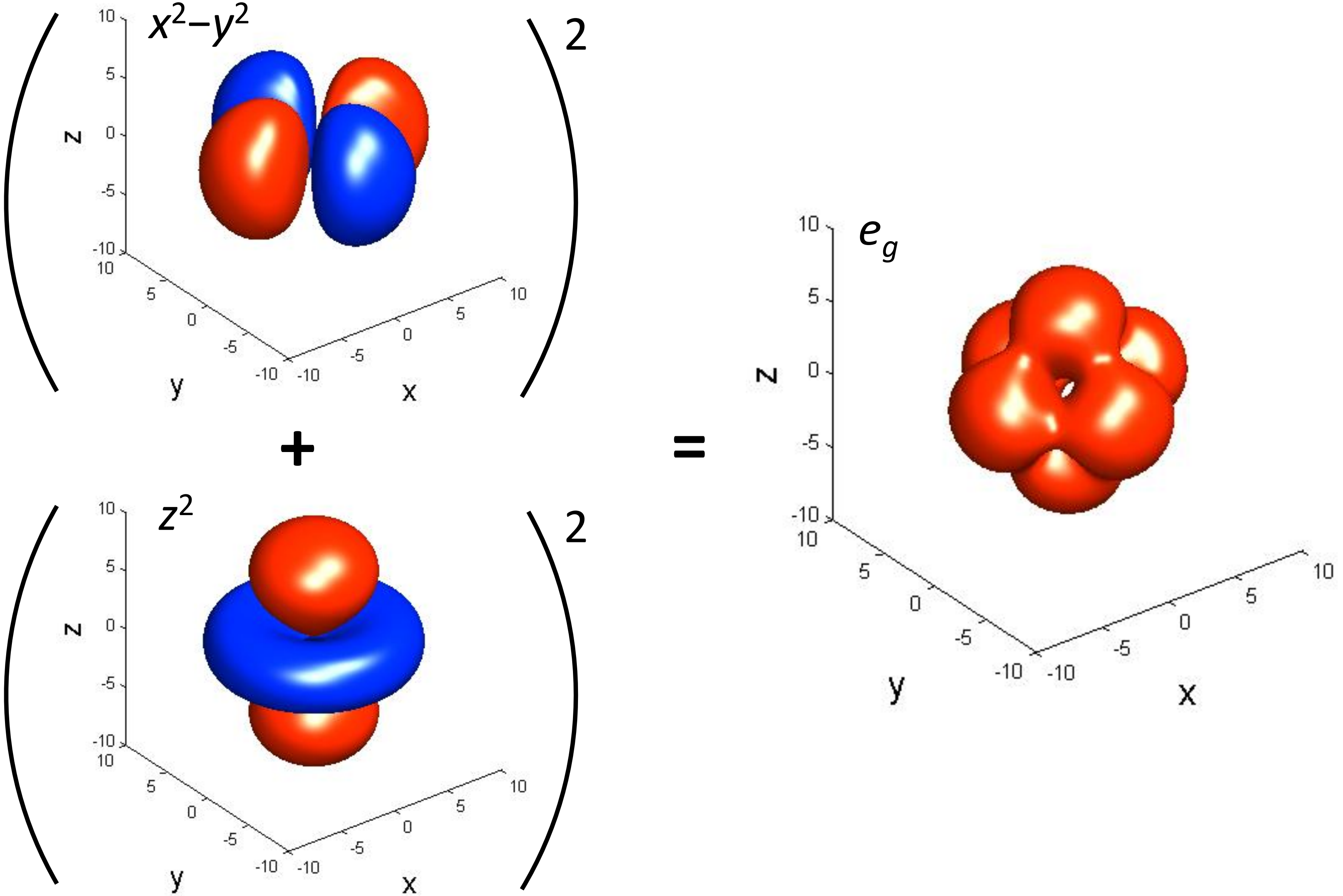
Octahedral crystal field



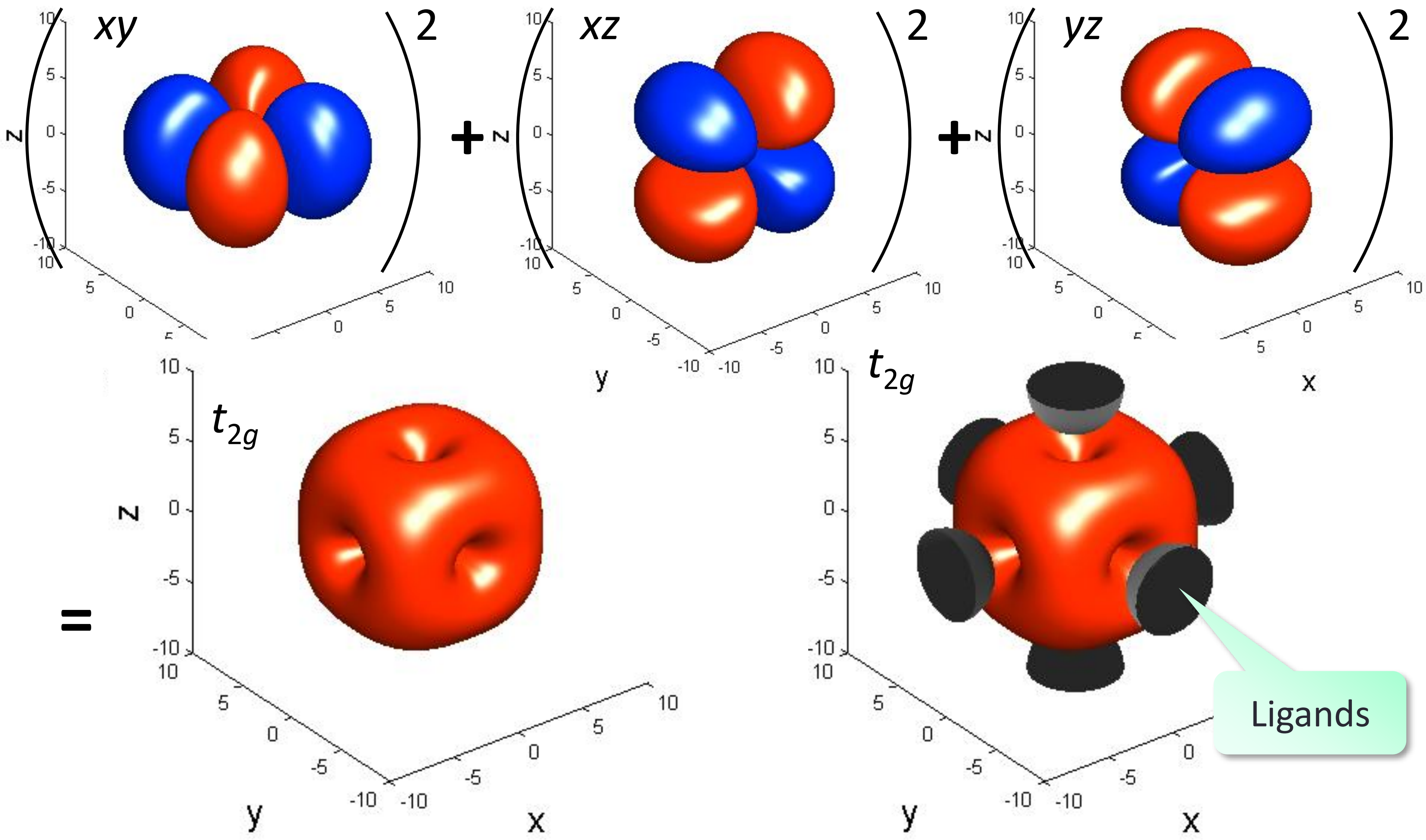
Large repulsion

Small repulsion

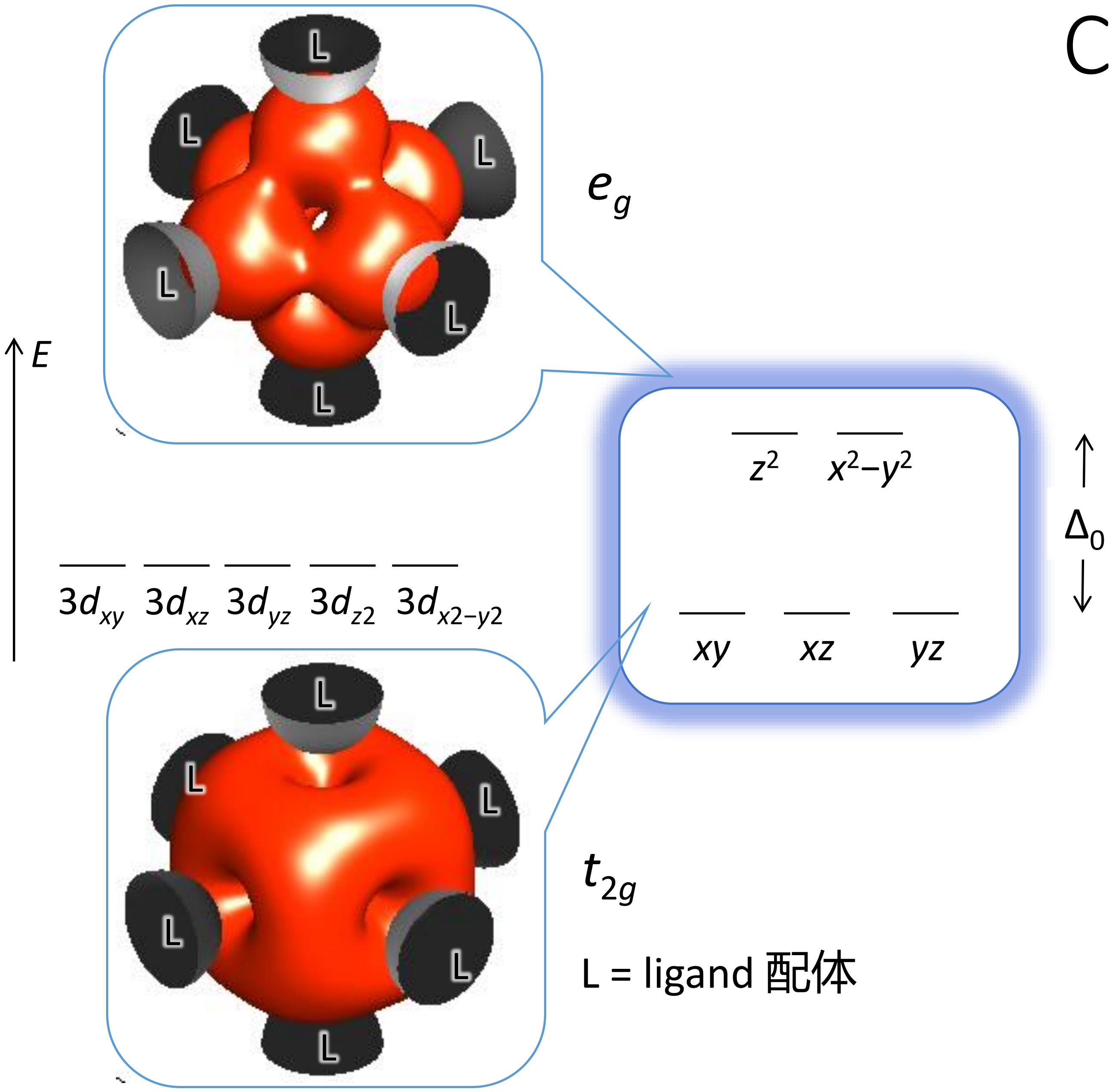
Which *d* Orbitals?



Which *d* Orbitals?



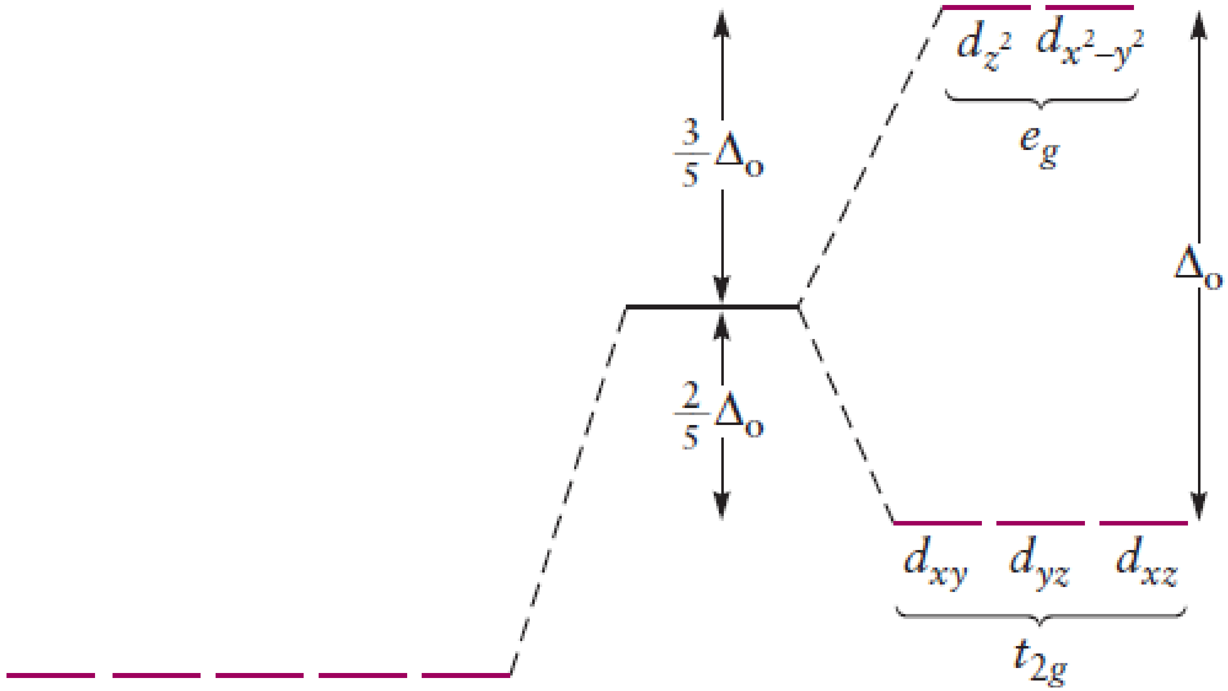
Crystal Field Theory



Crystal Field Splitting
晶体场分裂能

L = ligand 配体

Crystal Field Theory: Optical and Magnetic Properties



Octahedral crystal field

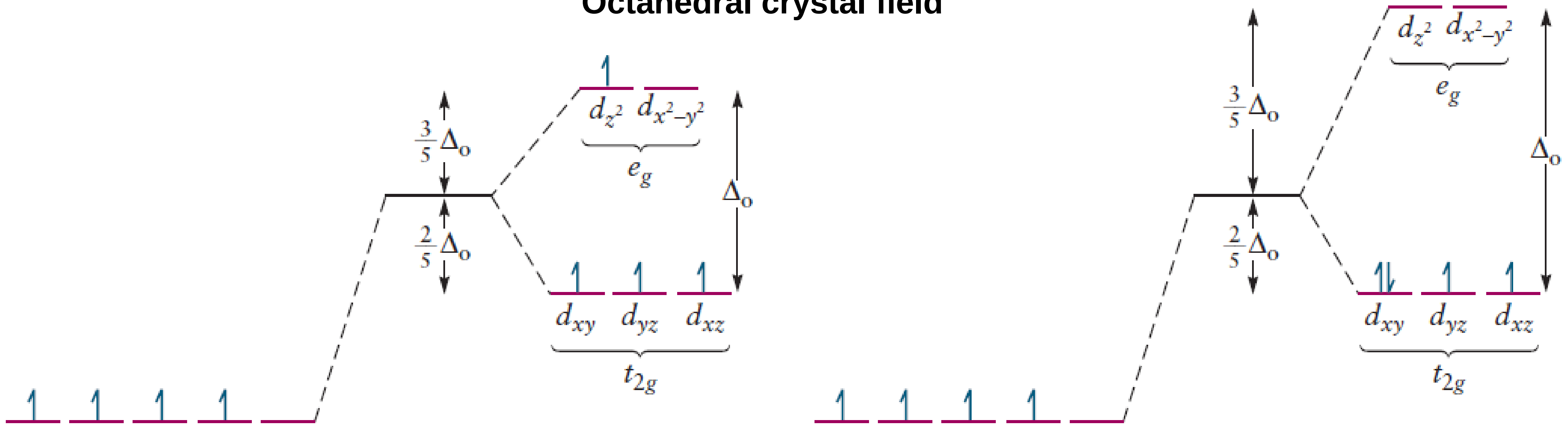
Δ_o: crystal field splitting energy

In an octahedral field, the energy of the three t_{2g} orbitals is lowered by 2/5 Δ_o, and that of the two e_g orbitals is raised by 3/5 Δ_o relative to the spherical field.

crystal field stabilization energies (CFSE): energy difference between electrons in an octahedral crystal field and those in a hypothetical spherical crystal field.

Crystal Field Theory: Optical and Magnetic Properties

Octahedral crystal field



(a) $[Mn(OH_2)_6]^{3+}$ (high spin) (b) $[Mn(CN)_6]^{3-}$ (low spin)

FIGURE 8.18 Electron configuration for (a) high-spin (Δ_o) and (b) low-spin (δ_o) Mn(III) complexes.

Δ_o is small (weak-field ligands):

- **high-spin configurations**

Δ_o is large (strong-field ligands)

- **low-spin configurations**

Electrons are paired in the lower energy t_{2g} orbital, and the e_g orbitals are not occupied until the t_{2g} levels are filled.

Crystal Field Theory: Optical and Magnetic Properties

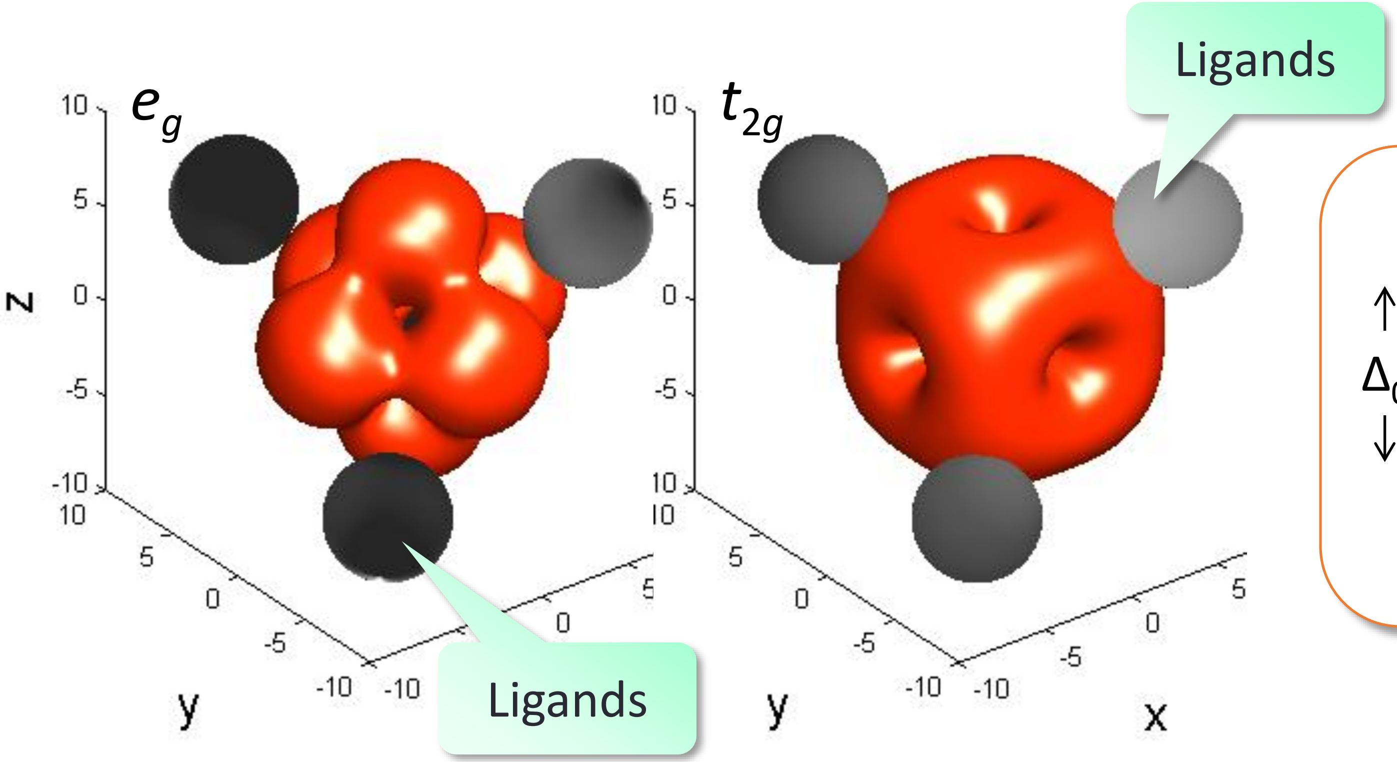
TABLE 8.5

Electron Configurations and Crystal Field Stabilization Energies for High- and Low-Spin Octahedral Complexes

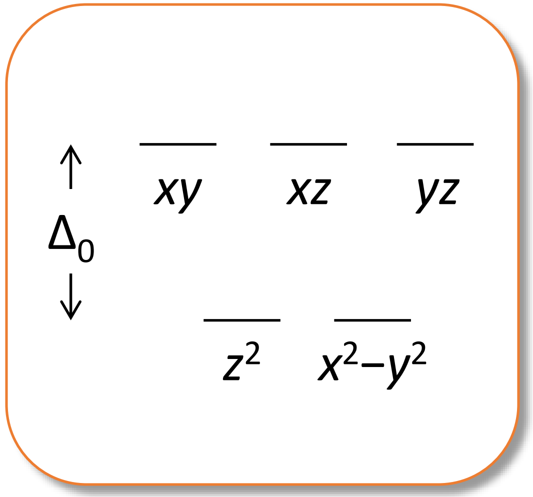
Configuration		d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
Examples		Ti^{3+}	Ti^{2+}, V^{3+}	V^{2+}, Cr^{3+}	Cr^{2+}, Mn^{3+}	Mn^{2+}, Fe^{3+}	Fe^{2+}, Co^{3+}	Co^{2+}, Ni^{3+}	Ni^{2+}, Pt^{2+}	Cu^{2+}	Zn^{2+}
HIGH SPIN	e_g	— —	— —	— —	↑ —	↑ ↑	↑ ↑	↑ ↑	↑ ↑	↑↓ ↑	↑↓ ↑↓
	t_{2g}	↑ — —	↑ ↑ —	↑ ↑ ↑	↑ ↑ ↑	↑ ↑ ↑	↑↓ ↑ ↑	↑↓ ↑↓ ↑	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓
	CFSE	$-\frac{2}{5}\Delta_o$	$-\frac{4}{5}\Delta_o$	$-\frac{6}{5}\Delta_o$	$-\frac{3}{5}\Delta_o$	0	$-\frac{2}{5}\Delta_o$	$-\frac{4}{5}\Delta_o$	$-\frac{6}{5}\Delta_o$	$-\frac{3}{5}\Delta_o$	0
LOW SPIN	e_g				— —	— —	— —	↑ —			
	t_{2g}				↑↓ ↑ ↑	↑↓ ↑↓ ↑	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓			
	CFSE	Same as high spin			$-\frac{8}{5}\Delta_o$	$-\frac{10}{5}\Delta_o$	$-\frac{12}{5}\Delta_o$	$-\frac{9}{5}\Delta_o$	Same as high spin		

CFSE, Crystal field stabilization energies.

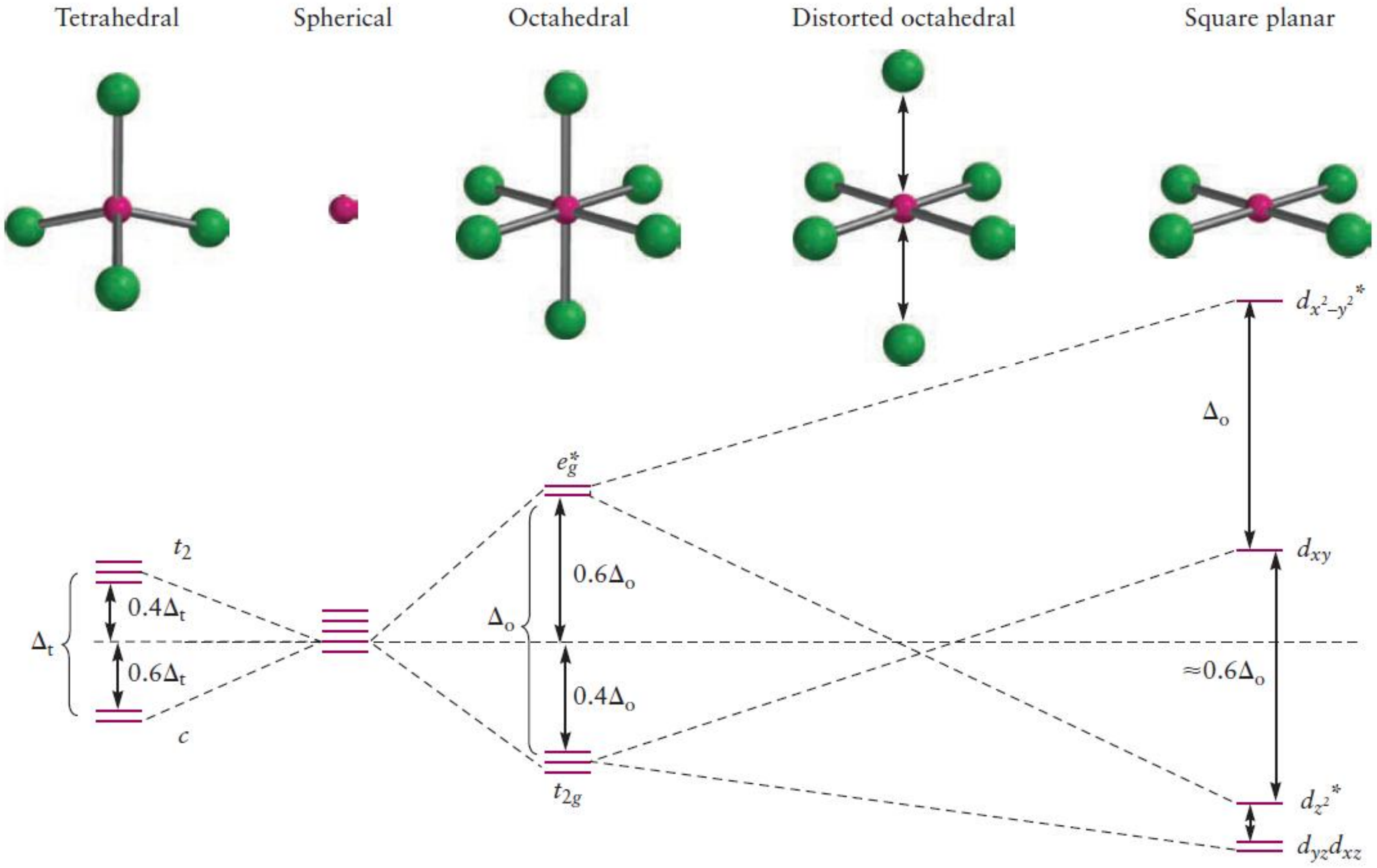
Tetrahedral Crystal Field



Lower energy Higher energy



Crystal Field Theory: summary



Tetrahedral: less stable

- The crystal field splitting is smaller
- The lower-energy set of levels is only doubly degenerate

Octahedral: most common:

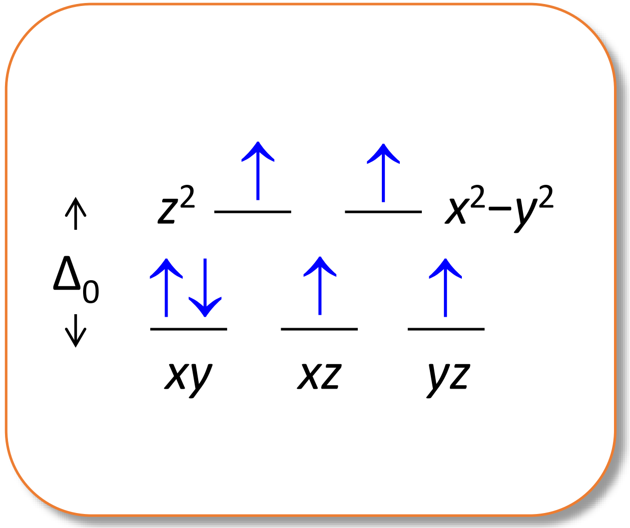
- the formation of 6 bonds, rather than 4, confers greater stability.

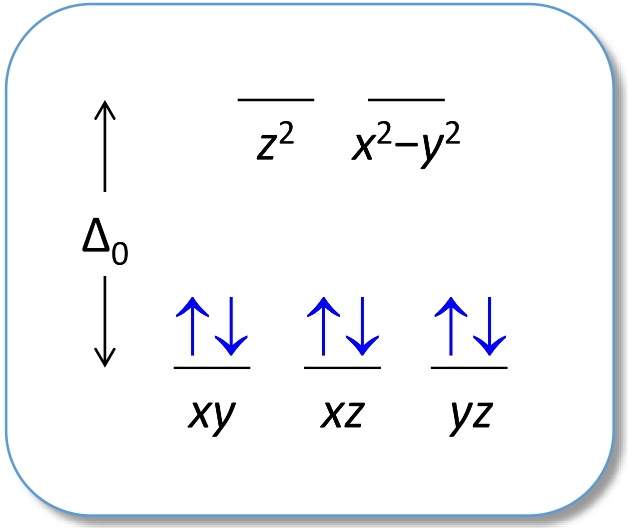
Square-planar: d^8 ions & strong field ligands.

- The low-spin configuration leaves the high-energy d_{x2-y2} orbital vacant is the most stable.

Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o





$$\Delta_o = h\nu = hc/\lambda_{max}$$

$[Co(NH_3)_6]^{3+}$	Orange
$[Co(NH_3)_4Cl_2]^{+}$	A green form and a violet form
$[Co(NH_3)_5(H_2O)]^{3+}$	Purple

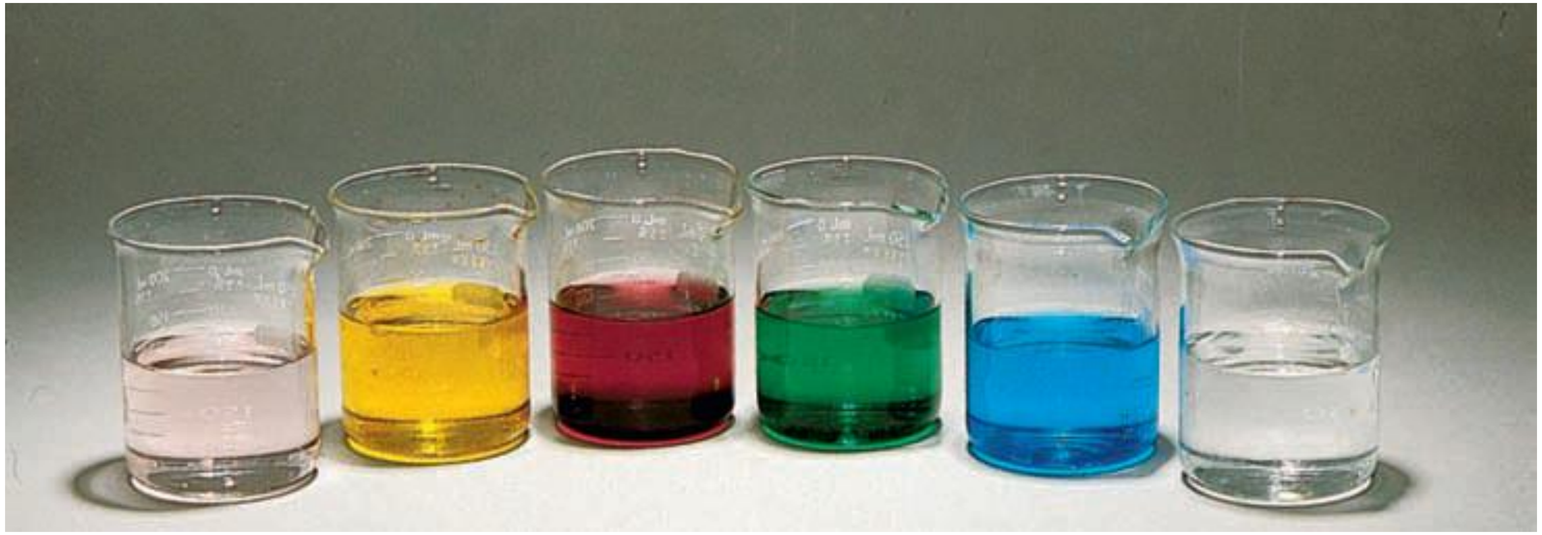
High spin

Low spin

$I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < CN^- \approx CO$

The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

Optical Properties and the Spectrochemical Series



d^{10} complexes (Zn^{2+} or Cd^{2+}) are colorless because all of the d levels (t_{2g} and e_g) are filled. There are no empty orbitals available to accept an excited electron, which means that the absorption is weak or nonexistent.

High-spin d^5 complexes such as $[Mn(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ also show weak absorption because promotion from a filled t_{2g} level to an empty e_g level would require a spin flip to satisfy the Pauli principle.

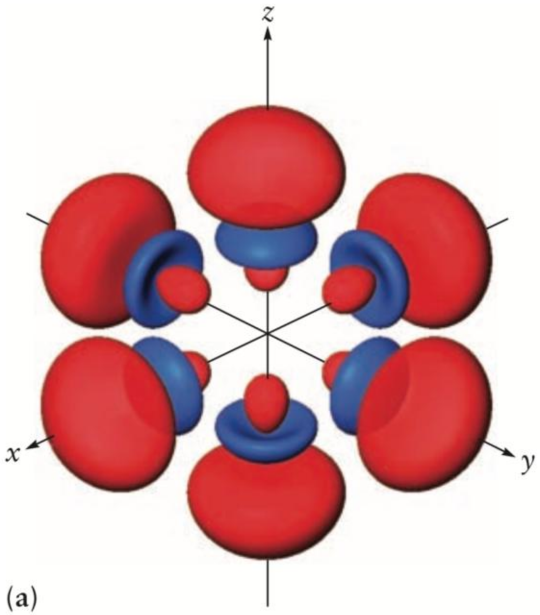
Bonding in Coordination Complexes: valence bond theory

Valence Bond Theory

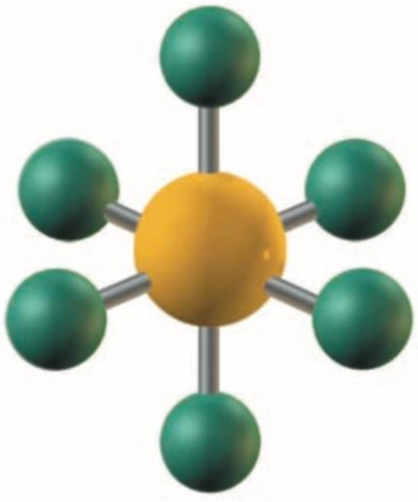
- Coordination complexes: a central atom surrounded by ligands in a symmetric arrangement, forming hybrid orbitals on the central atom with the appropriate symmetry to bond to these ligands

- Often, little interaction occurs among metal–ligand bonds, so the local description is reasonable.

Bonding in Coordination Complexes: valence bond theory



(a)


(b)

d^2sp^3 hybrid orbitals

$$\chi_1 = \sqrt{\frac{1}{6}}\,[s + \sqrt{3}\,(p_z) + \sqrt{2}\,(d_{z^2})]$$

$$\chi_2 = \sqrt{\frac{1}{6}}\left[s + \sqrt{3}\,(p_z) - \sqrt{\frac{1}{2}}\,(d_{z^2}) + \sqrt{\frac{3}{2}}\,(d_{x^2-y^2})\right]$$

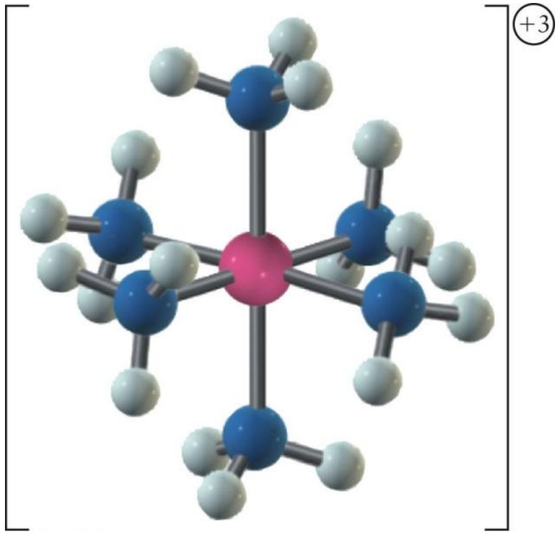
$$\chi_3 = \sqrt{\frac{1}{6}}\left[s + \sqrt{3}\,(p_z) - \sqrt{\frac{1}{2}}\,(d_{z^2}) - \sqrt{\frac{3}{2}}\,(d_{x^2-y^2})\right]$$

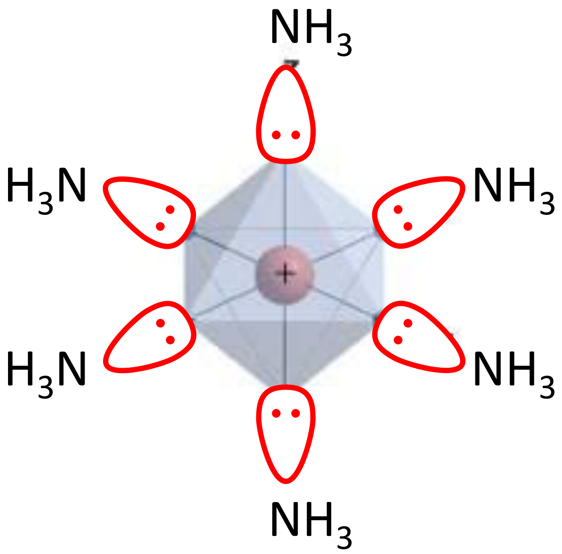
$$\chi_4 = \sqrt{\frac{1}{6}}\left[s - \sqrt{3}\,(p_z) - \sqrt{\frac{1}{2}}\,(d_{z^2}) + \sqrt{\frac{3}{2}}\,(d_{x^2-y^2})\right]$$

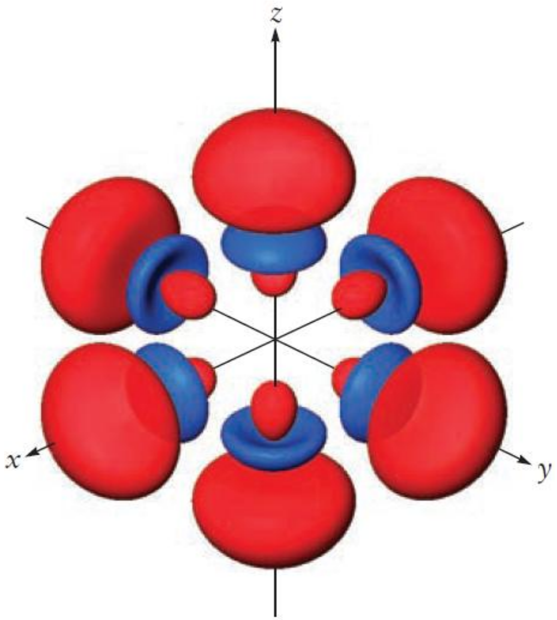
$$\chi_5 = \sqrt{\frac{1}{6}}\left[s - \sqrt{3}\,(p_z) - \sqrt{\frac{1}{2}}\,(d_{z^2}) - \sqrt{\frac{3}{2}}\,(d_{x^2-y^2})\right]$$

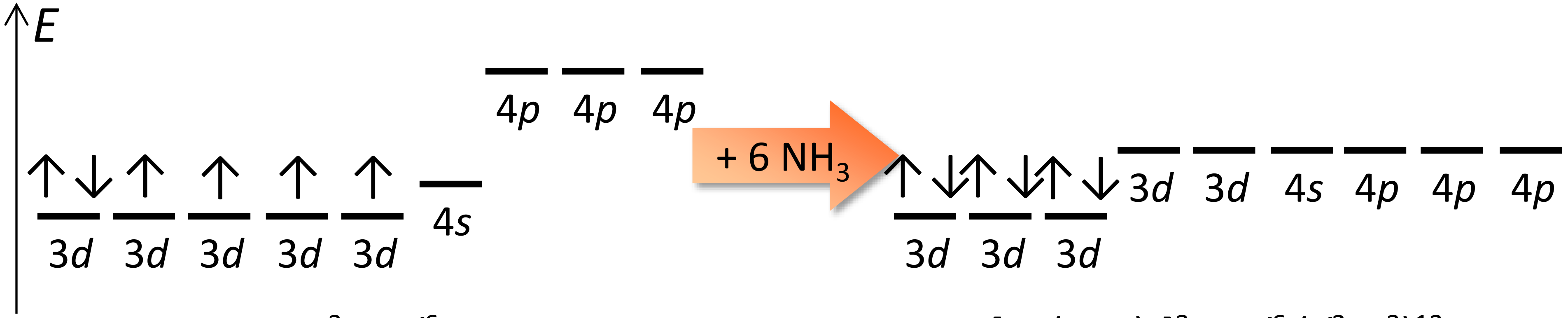
$$\chi_6 = \sqrt{\frac{1}{6}}\,[s - \sqrt{3}\,(p_z) + \sqrt{2}\,(d_{z^2})]$$

Orbital Hybridization in $[Co(NH_3)_6]^{3+}$









Co^{3+}: $3d^6$

$[Co(NH_3)_6]^{3+}$: $3d^6$ $(d^2sp^3)^{12}$

Bonding in Coordination Complexes: valence bond theory

TABLE 8.7

Examples of Hybrid Orbitals and Bonding in Complexes

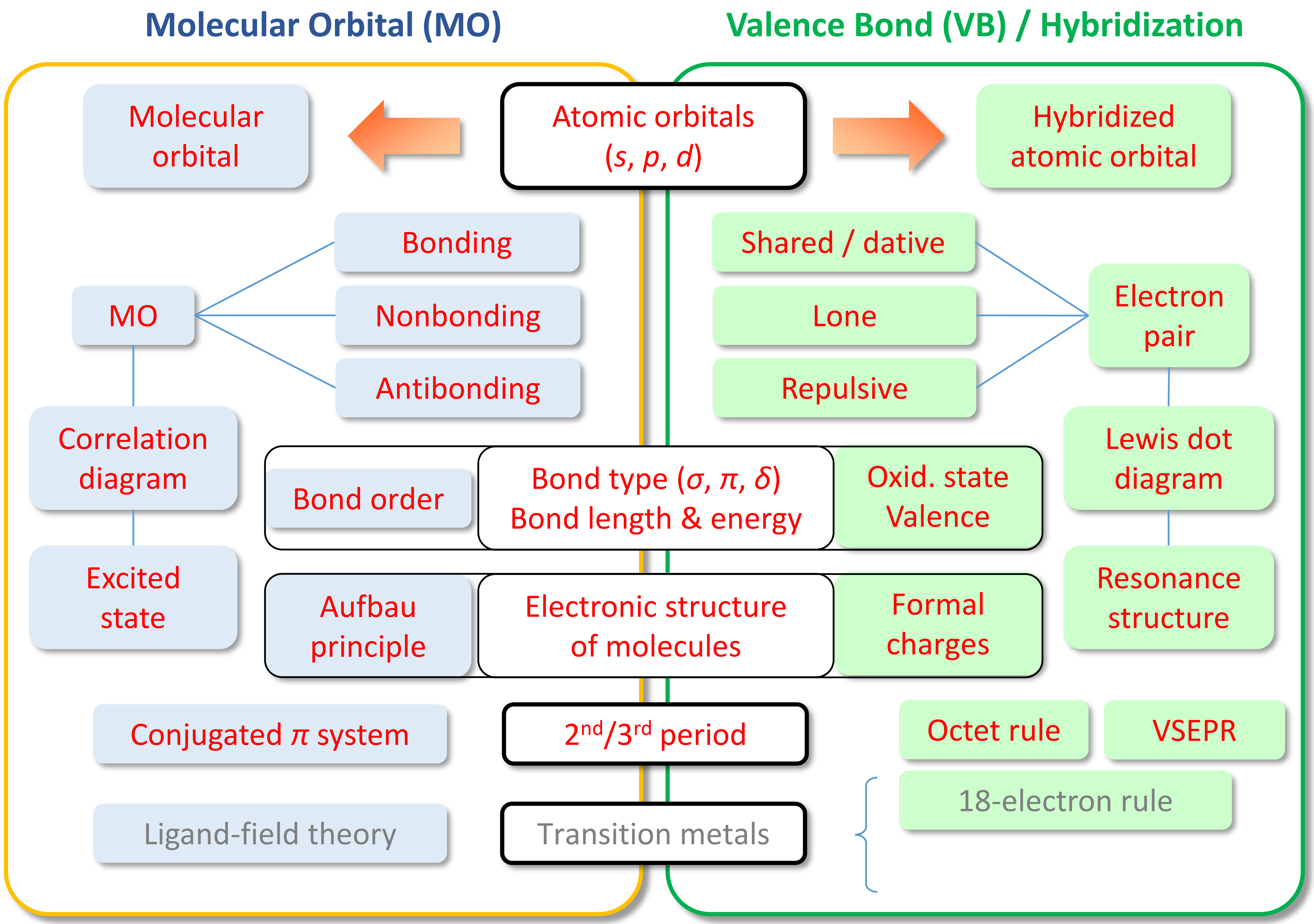
Coordination Number	Hybrid Orbital	Configuration	Examples
2	*sp*	Linear	$[Ag(NH_3)_2]^+$
3	sp^2	Trigonal planar	BF_3, NO_3^-, $[Ag(PR_3)_3]^+$
4	sp^3	Tetrahedral	$Ni(CO)_4$, $[MnO_4]^-$, $[Zn(NH_3)_4]^{2+}$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$
5	dsp^3	Trigonal bipyramidal	TaF_5, $[CuCl_5]^{3-}$, $[Ni(PEt_3)_2Br_3]$
6	d^2sp^3	Octahedral	$[Co(NH_3)_6]^{3+}$, $[PtCl_6]^{2-}$

From G.E. Kimball, Directed valence. *J. Chem. Phys.* 1940, 8, 188.

Major points of today's lecture

- **Bonding in transition metal compounds and coordination complexes**

1. **Chemistry of the Transition Metals**
2. **Introduction to Coordination Chemistry**
3. **Structures of Coordination Complexes**
4. **Crystal Field Theory: Optical and Magnetic Properties**
5. **Optical Properties and the Spectrochemical Series**
6. **Bonding in Coordination Complexes**


Molecular Orbital (MO)
Valence Bond (VB) / Hybridization
Molecular orbital
Atomic orbitals (s, p, d)
Hybridized atomic orbital
Bonding
Nonbonding
Antibonding
MO
Shared / dative
Lone
Repulsive
Electron pair
Correlation diagram
Bond order
Bond type (σ, π, δ)
Bond length & energy
Oxid. state
Valence
Lewis dot diagram
Excited state
Aufbau principle
Electronic structure of molecules
Formal charges
Resonance structure
Conjugated π system
2nd/3rd period
Octet rule
VSEPR
18-electron rule
Ligand-field theory
Transition metals

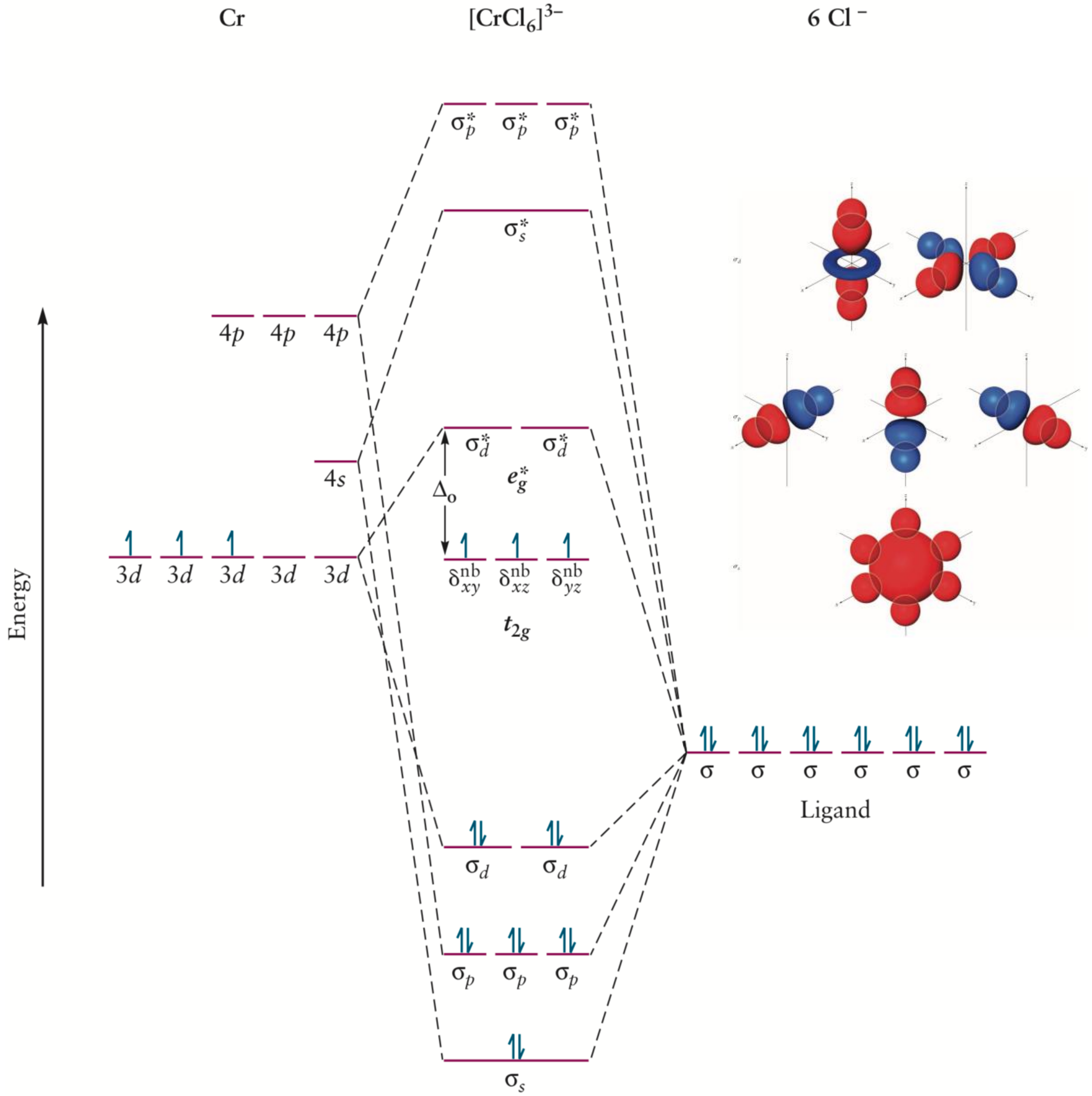
Bonding in Coordination Complexes: Molecular Orbital Theory

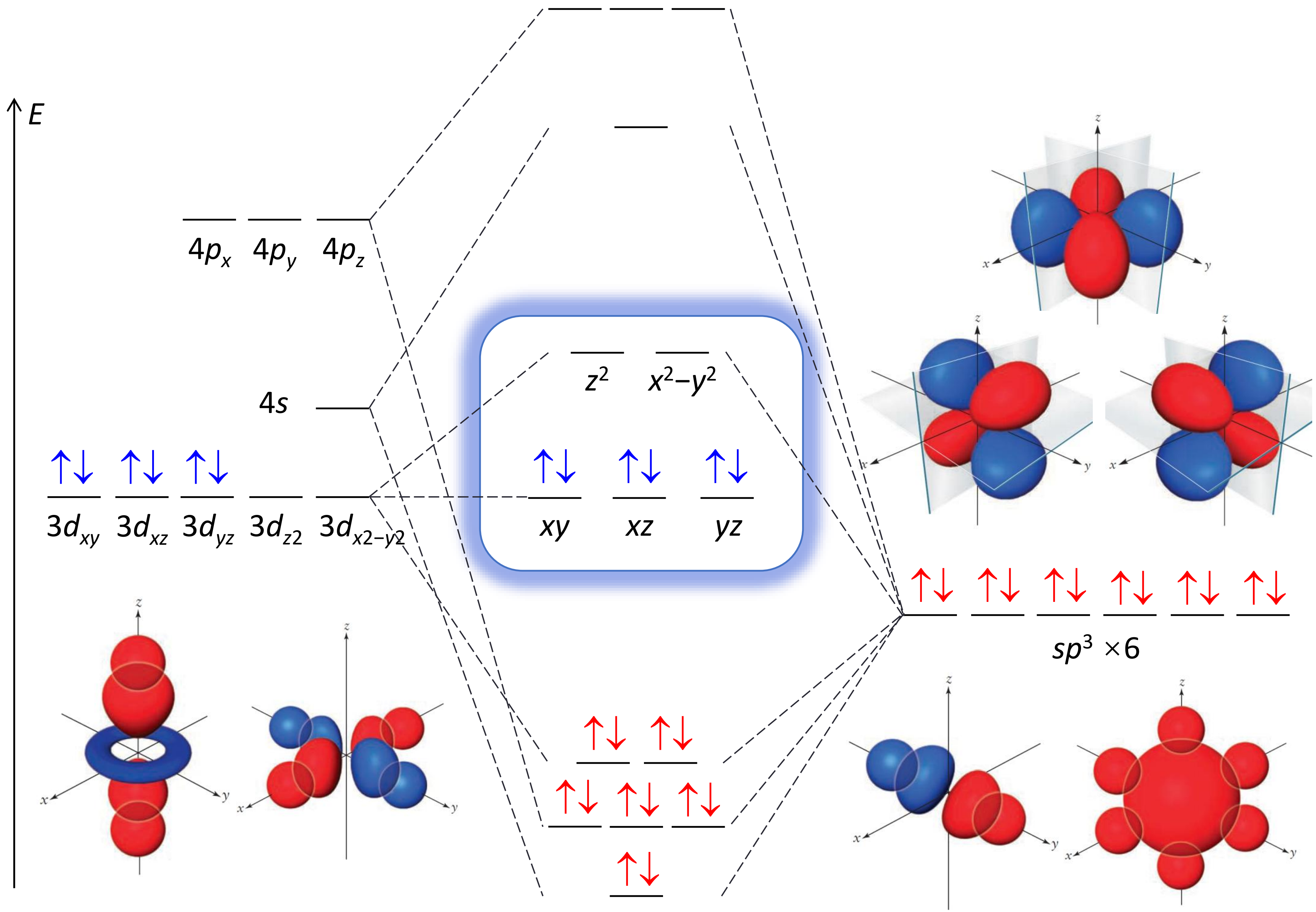
Molecular Orbital Theory

- **Metal orbitals: 3d, 4s, and 4p orbitals.**
- **Ligand orbitals: sp^3 orbitals containing a lone pair electron in H_2O, or p orbitals from F^- ions directed toward the metal ion.**

The d_{xy}, d_{yz}, and d_{xz} orbitals, whose lobes are oriented at 45 degrees to the bond axes, do not overlap with the ligand orbitals and are nonbonding.

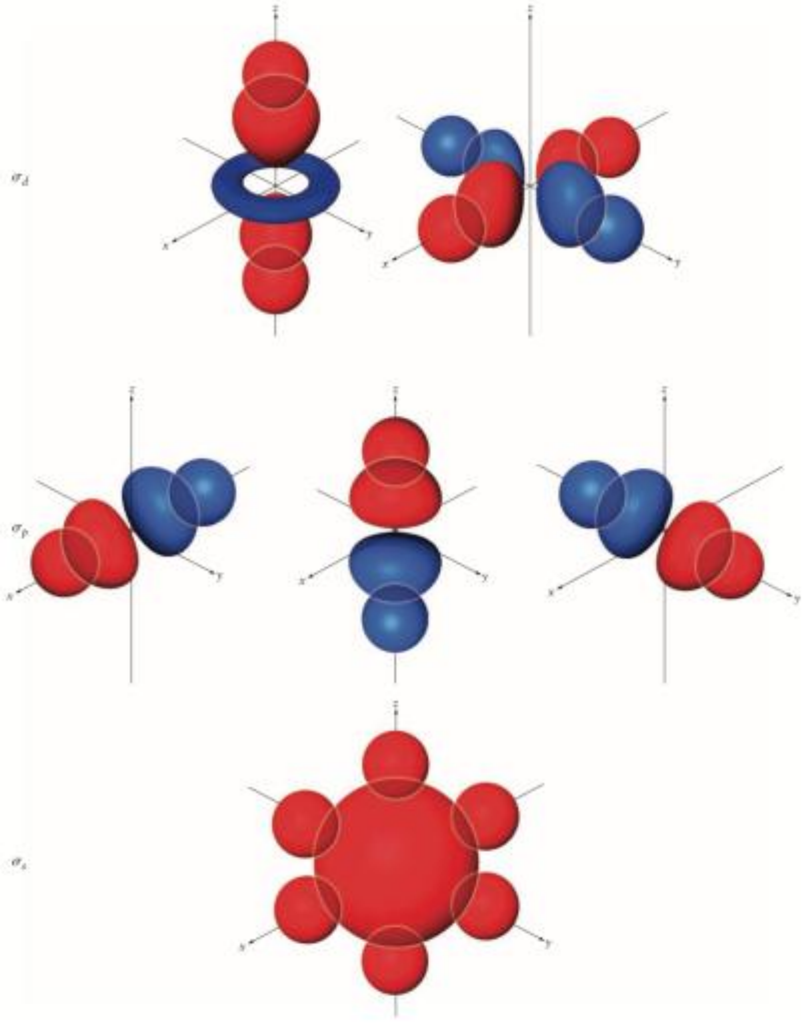
Ligand-to-metal (L → M) σ donation





E
4p_x 4p_y 4p_z
4s
3d_xy 3d_xz 3d_yz 3d_z2 3d_x2−y2
z^2 x^2−y^2
xy xz yz
sp^3 ×6

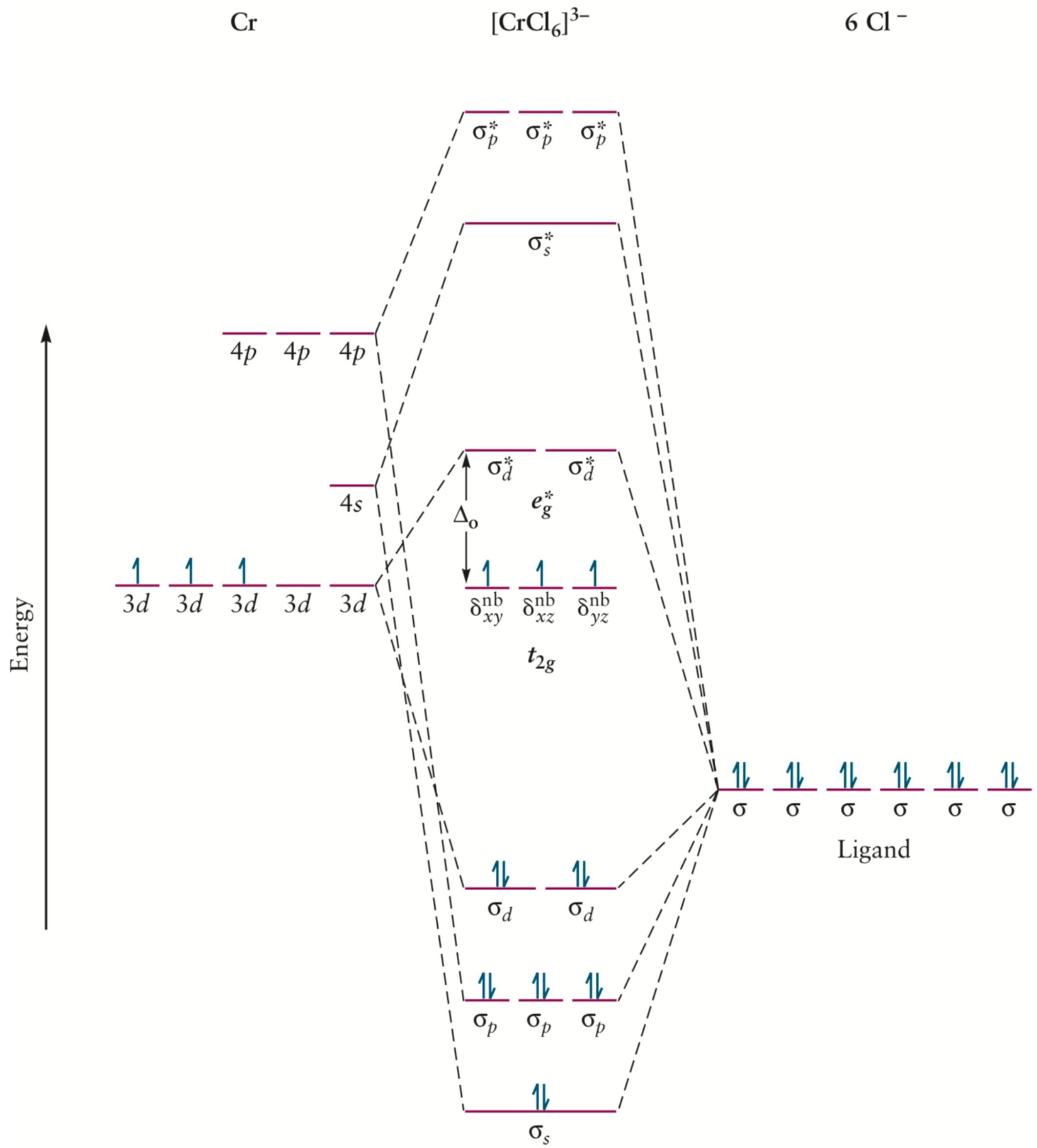
Bonding in Coordination Complexes: Molecular Orbital Theory



σ_d orbital derived from the metal d_{z2} orbital delocalized along the z-axis and the metal d_{x2-y2} orbital delocalized over the x-y plane.

σ_p orbitals are each delocalized along one of the Cartesian axes.

σ_s orbital is delocalized over the entire complex.



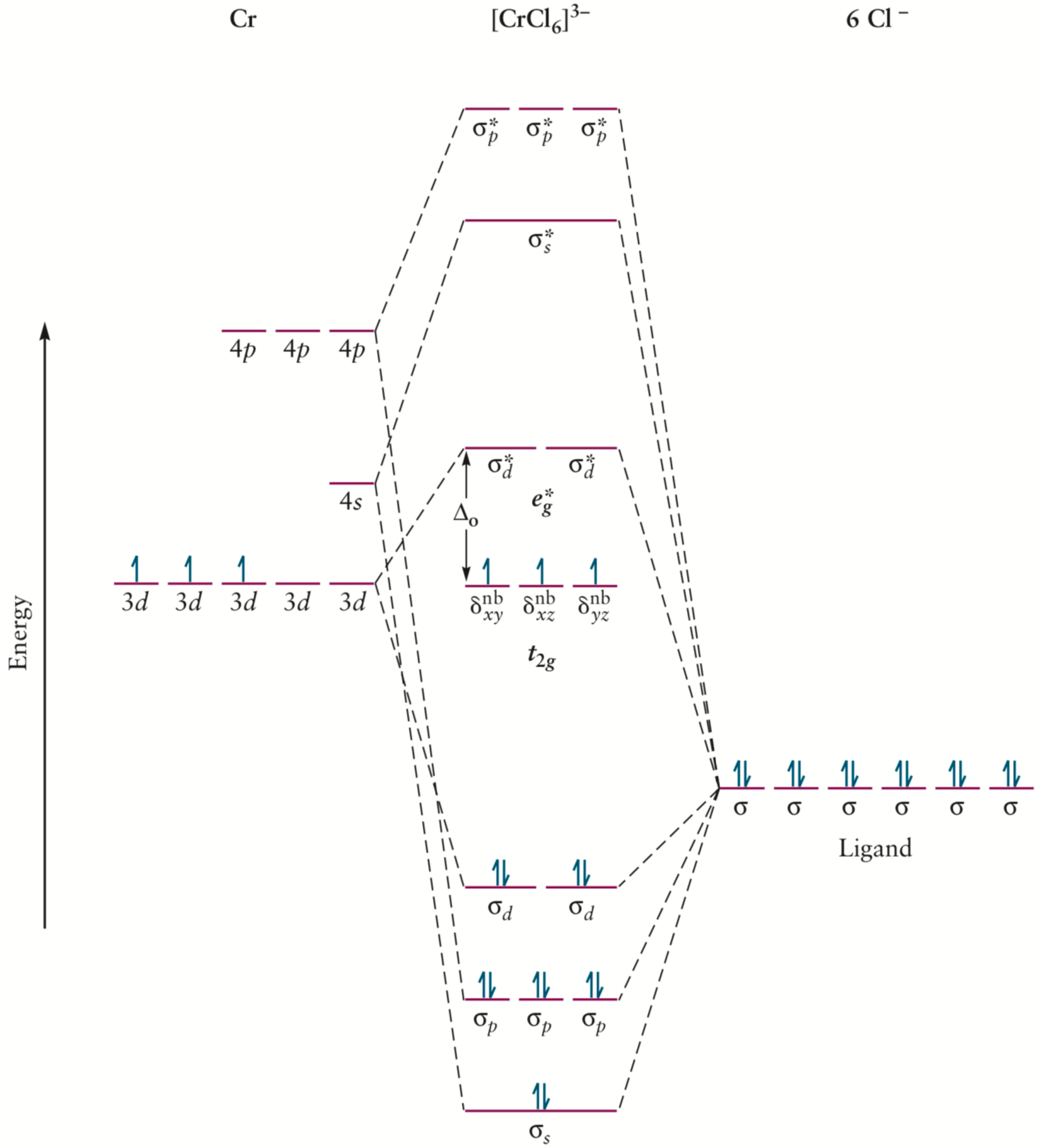
Ligand-to-metal (L → M) σ donation

Bonding in Coordination Complexes: Molecular Orbital Theory

Ligand-to-metal (L → M) σ donation

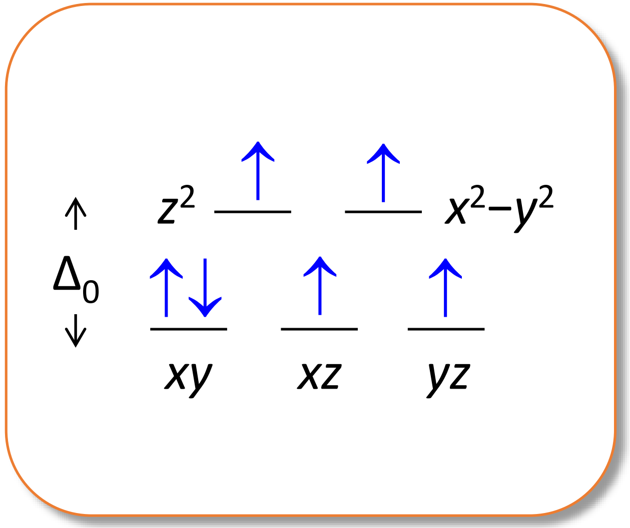
Strong interactions decrease the energy of (stabilize) the bonding σ_d orbital but increase the energy of (de-stabilize) the σ^*_d orbital.

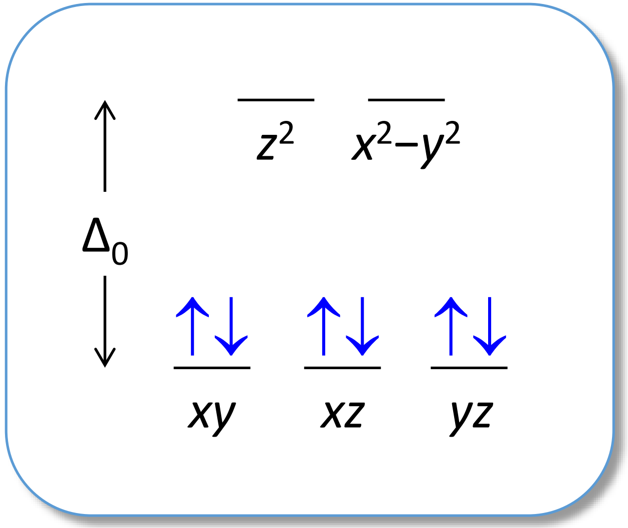
This description explains the increase in Δ_o as the strength of the metal–ligand interaction increases.



Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o





$$\Delta_o = h\nu = hc/\lambda_{max}$$

$[Co(NH_3)_6]^{3+}$	Orange
$[Co(NH_3)_4Cl_2]^{+}$	A green form and a violet form
$[Co(NH_3)_5(H_2O)]^{3+}$	Purple

High spin

Low spin

Weak-field ligands

Strong-field ligands

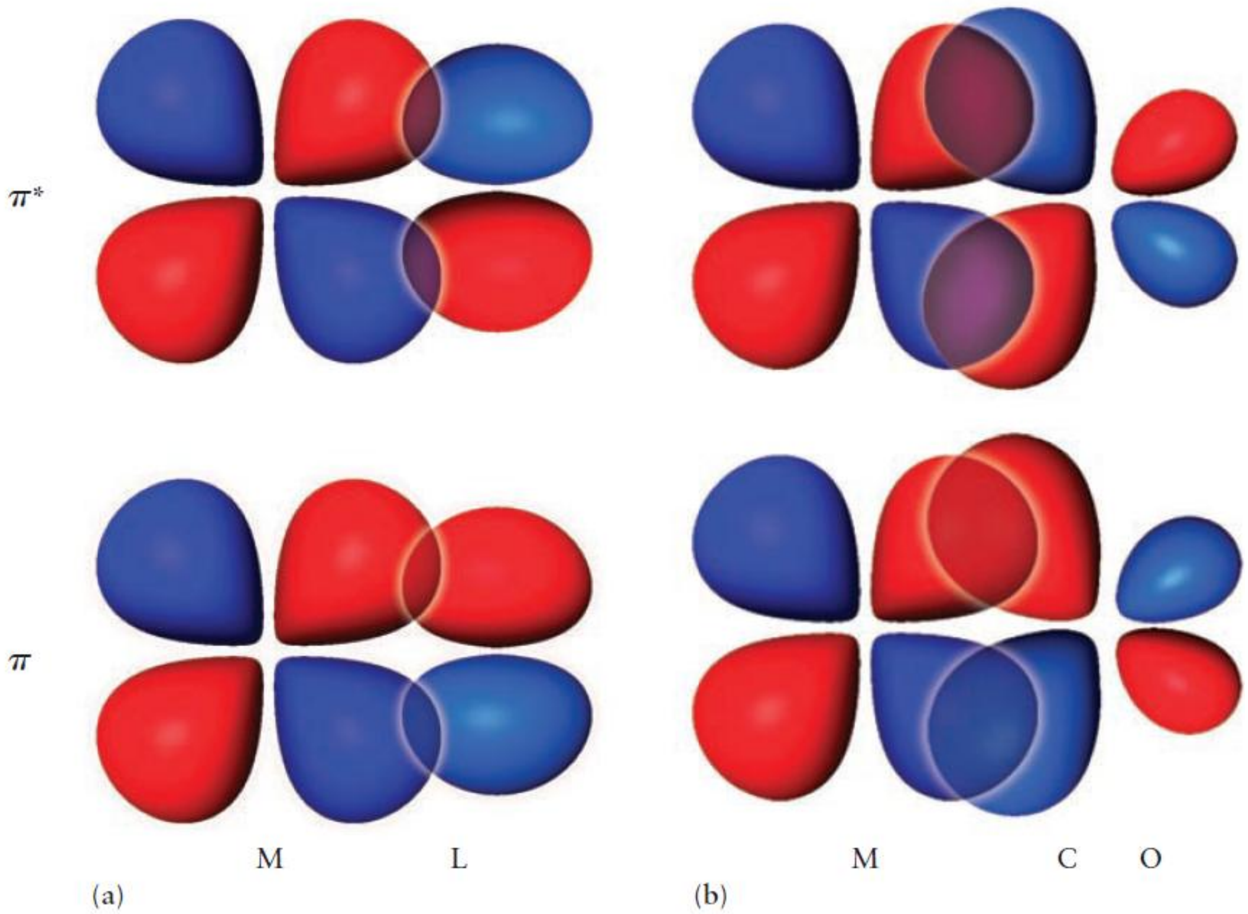
$I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < CN^- \approx CO$

The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

Bonding in Coordination Complexes: Molecular Orbital Theory

The different strengths of ligands such as the halides and those containing unfilled antibonding π orbitals like CO and CN^- cannot be explained considering σ bonding alone.

FIGURE 8.29 (a) Bonding and antibonding π orbitals formed by "side-by-side" overlap of a metal *d* orbital with an atomic ligand *p* orbital. (b) Bonding and antibonding π orbitals formed by "side-by-side" overlap of a metal *d* orbital with a molecular ligand π^* orbital.



Heteronuclear Diatomic Molecules

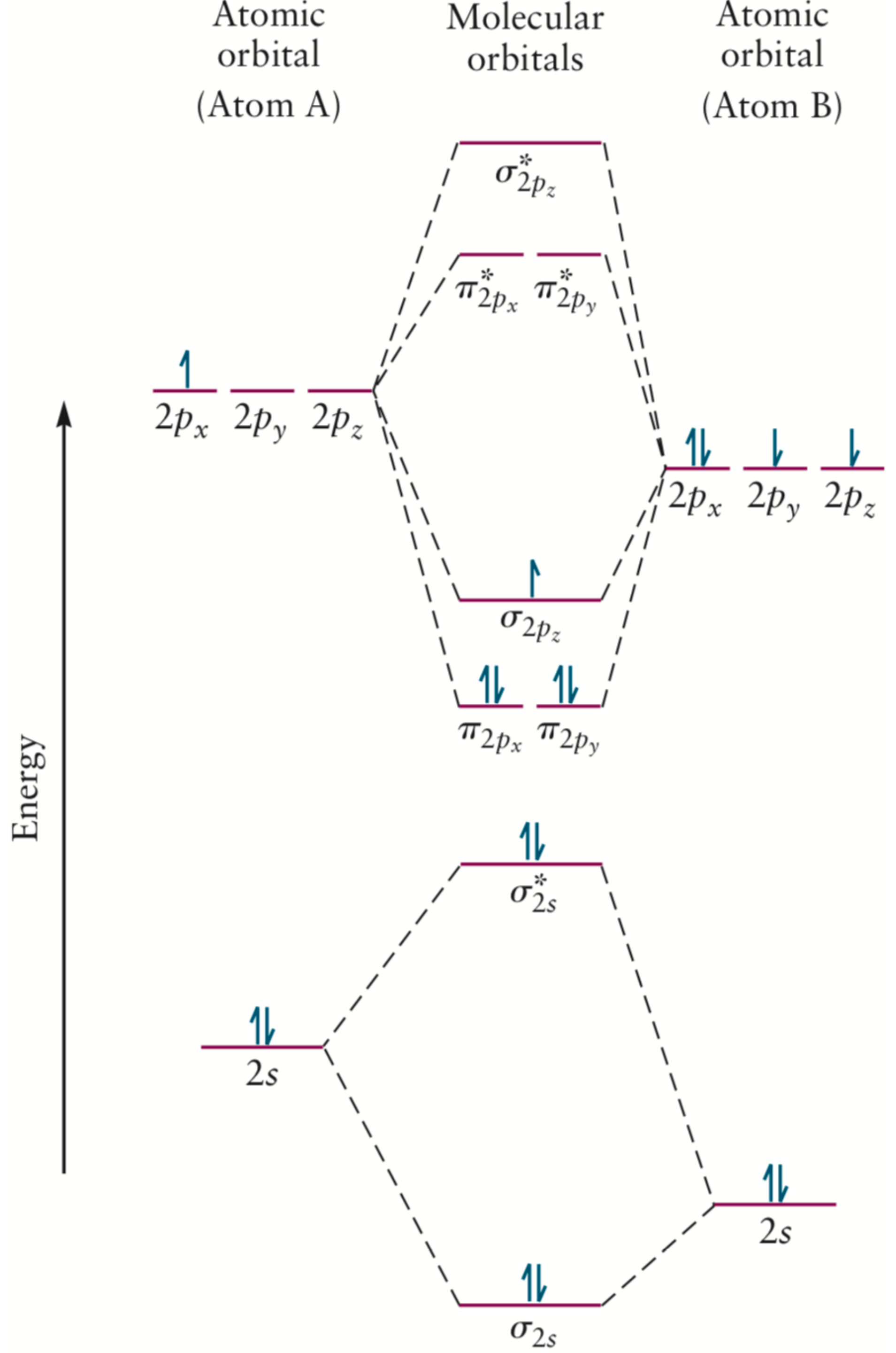


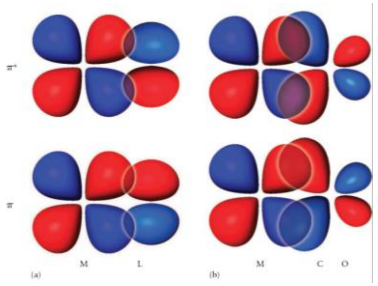
FIGURE 6.20 Correlation diagram for heteronuclear diatomic molecules, AB. The atomic orbitals for the more electronegative atom (B) are displaced downward because they have lower energies than those for A. The orbital filling shown is that for (boron monoxide) BO.

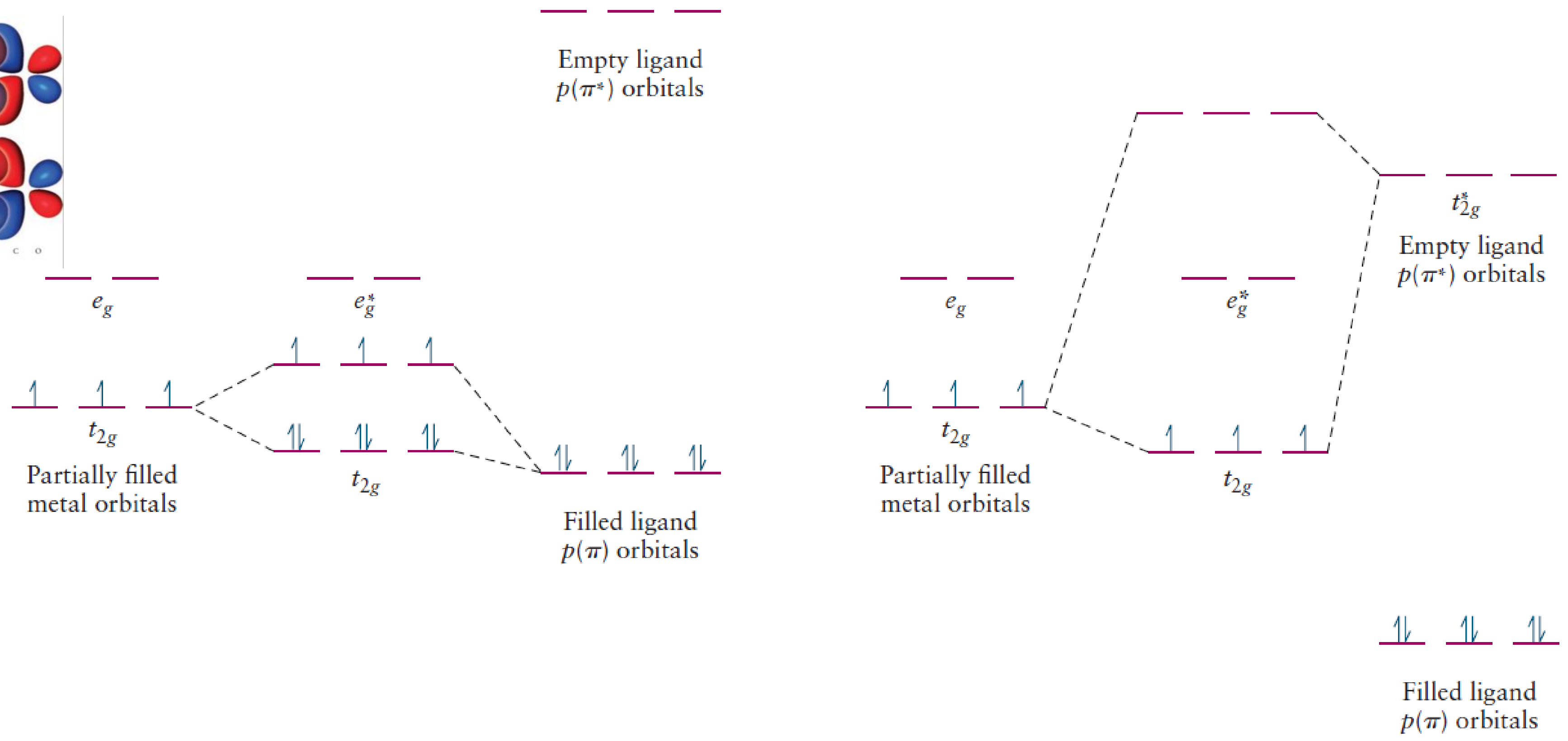
L→M π donation:

- metals in high oxidation states (few d electrons, empty d orbitals)
- ligands with filled orbitals whose energies lie close to those of the empty d orbitals (the halides, water, OH and ammonia)

M→L π backbonding

- metals in low oxidation states (many d electrons, few unfilled d orbitals)
- ligands with empty orbitals whose energies lie close to those of the filled d orbitals (d orbitals of P or S, or π^* orbitals in ligands such as CO and CN)





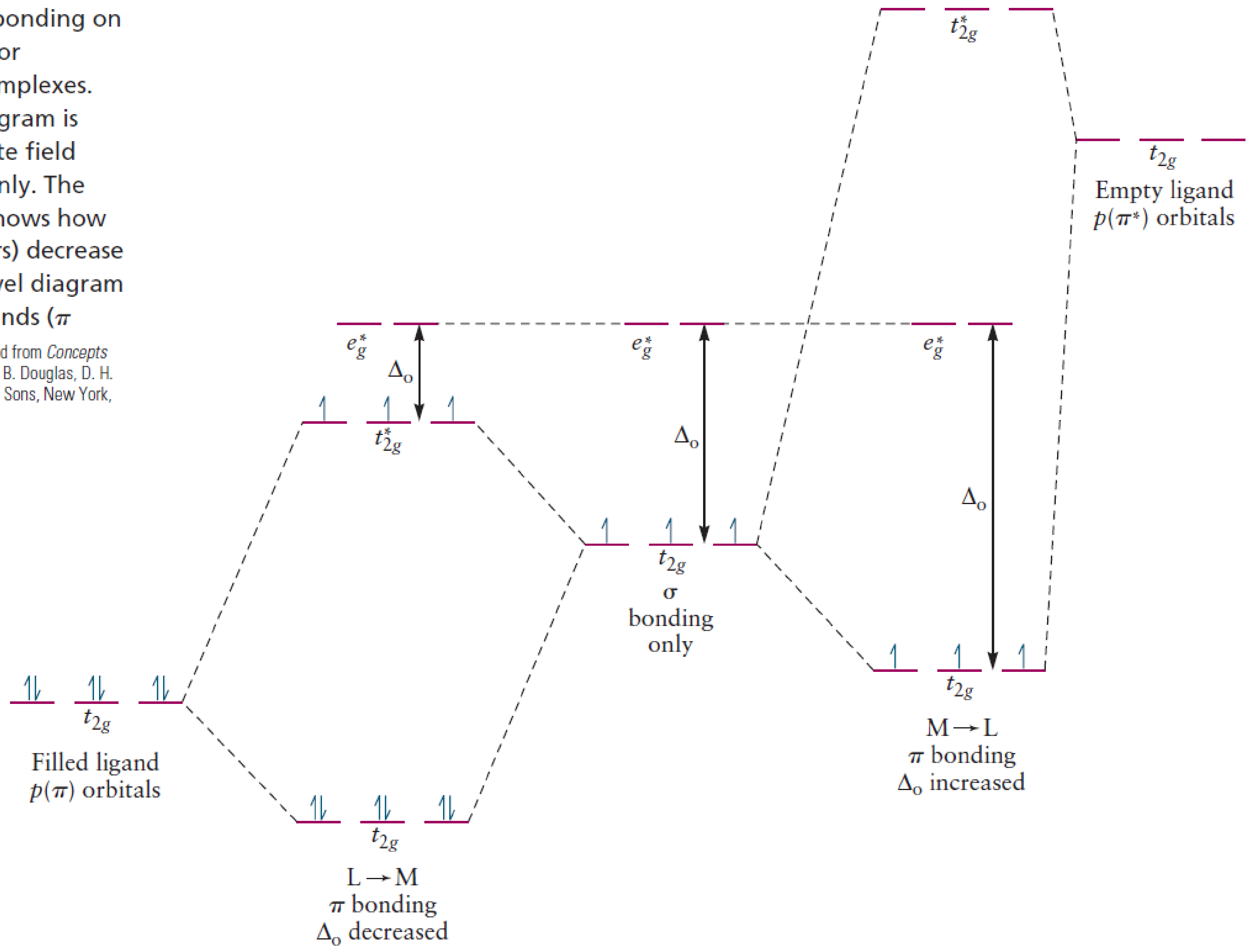
(a) π donor ligands (b) π acceptor ligands

$I^- < Br^- < Cl^- < F^- < OH_2 < NH_3 < en < CO, CN^-$

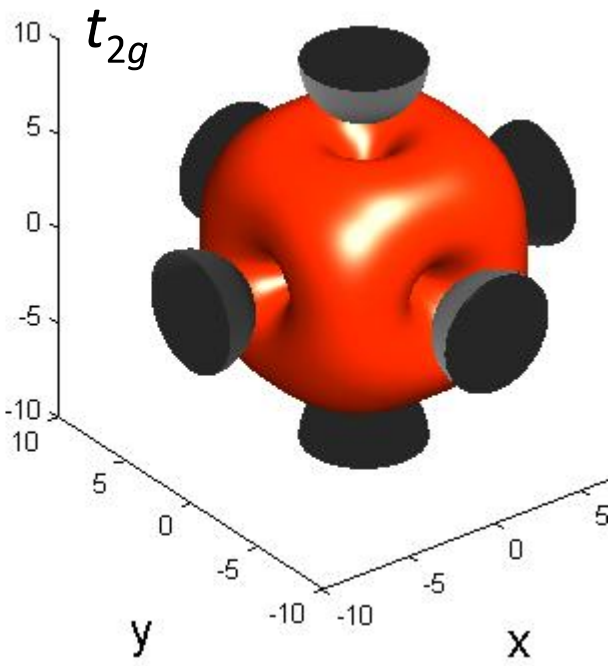
π donors σ donors π acceptors

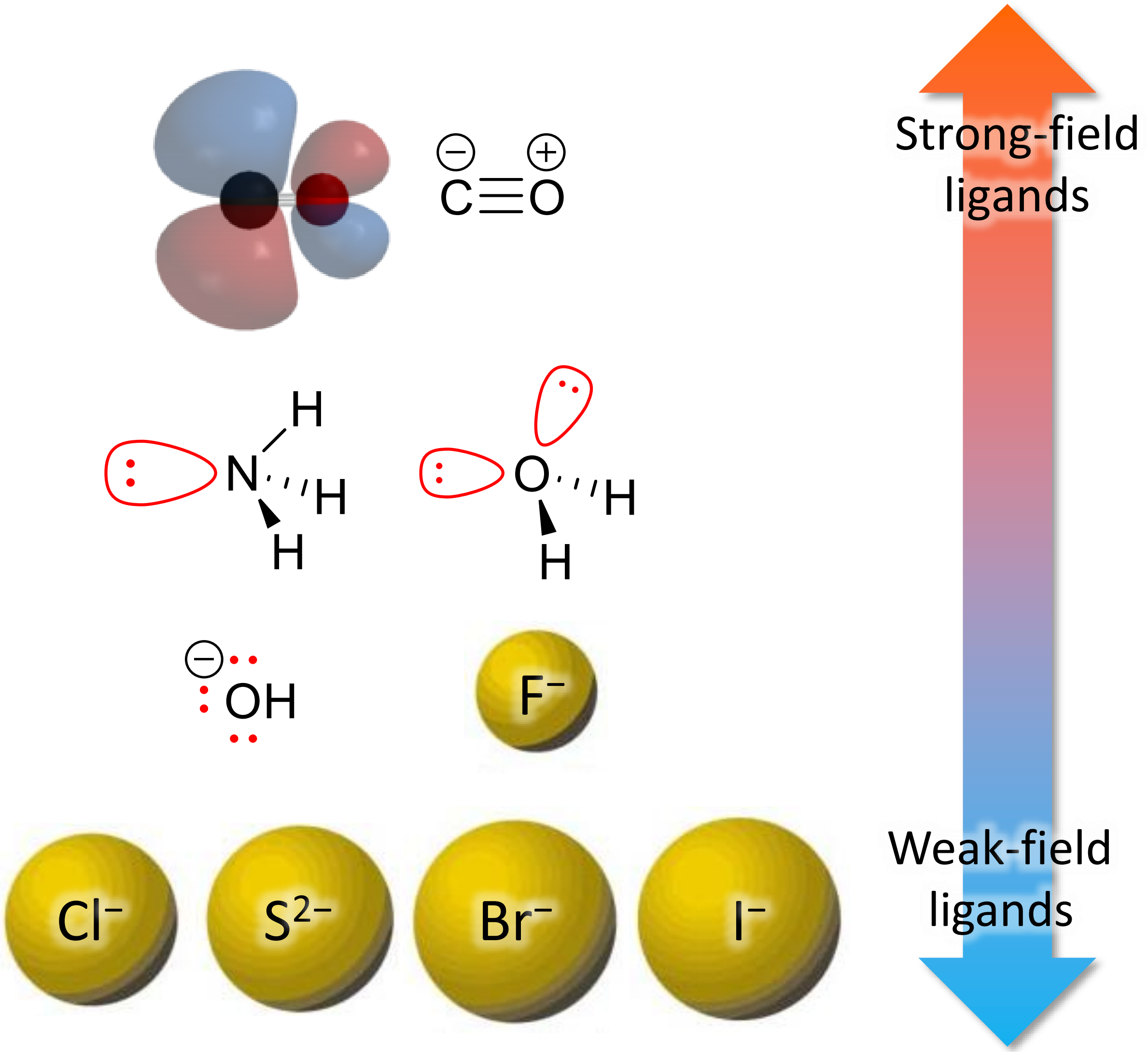
Bonding in Coordination Complexes: Molecular Orbital Theory

FIGURE 8.32 Effect of π bonding on the energy-level structure for octahedral coordination complexes. The center energy-level diagram is appropriate for intermediate field ligands that are σ donors only. The left energy-level diagram shows how weak field ligands (π donors) decrease Δ_o, and the right energy-level diagram shows how strong field ligands (π acceptors) increase Δ_o. (Adapted from *Concepts and Models of Inorganic Chemistry*, 2nd edition, B. Douglas, D. H. McDaniel, and J. J. Alexander, John Wiley and Sons, New York, 1983, p. 293.)

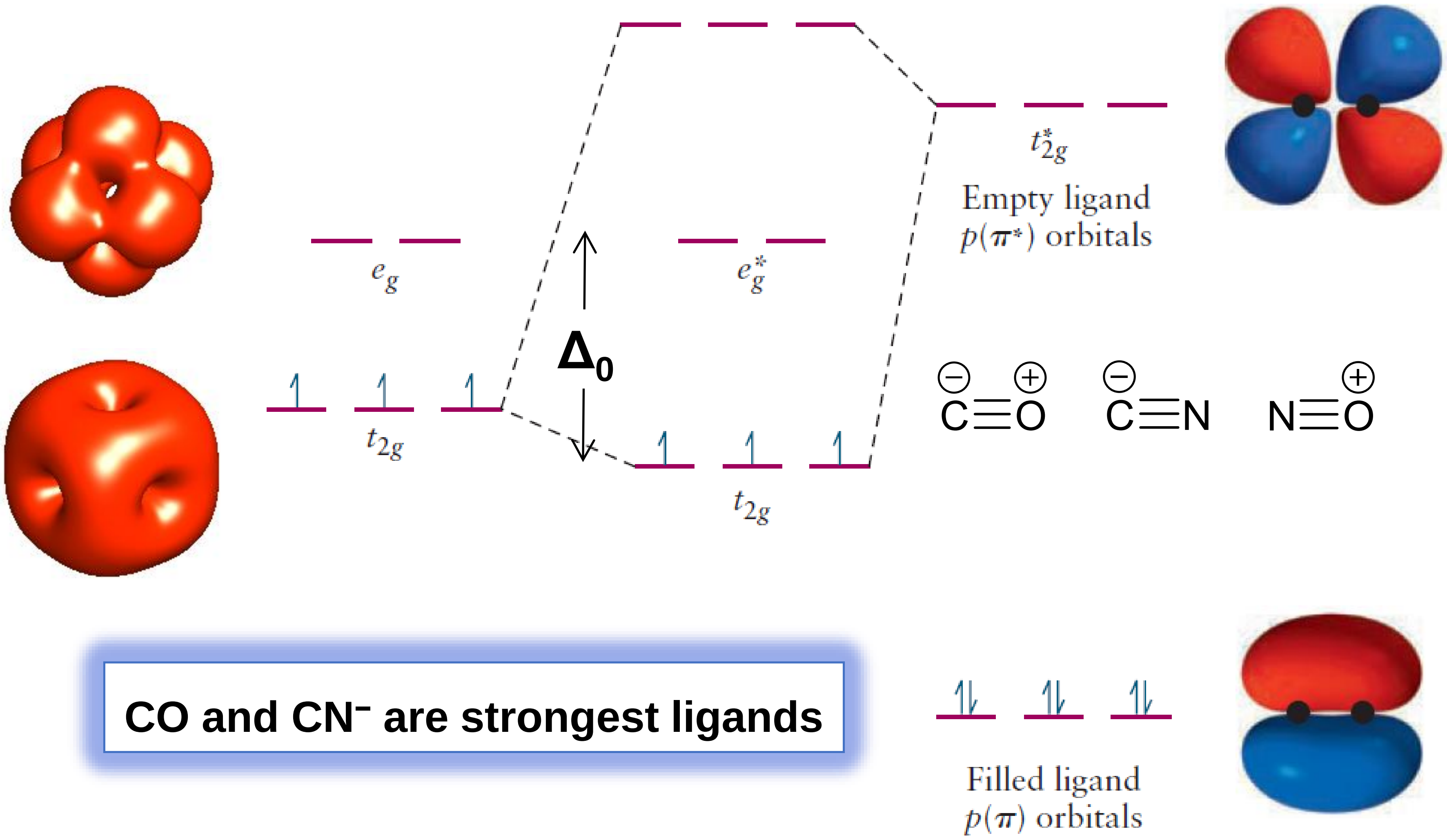


Optical Properties and the Spectrochemical Series





Strongest Ligands: π Acceptors



CO and CN^- are strongest ligands

Construct an MO correlation diagram for the complex ion $[Ni(NH_3)_6]^{2+}$ and populate the orbitals with electrons. Do you expect π bonding to be important for this complex?

What features of this diagram correspond to those predicted using crystal field theory?

Solution

The correlation diagram looks like that shown in Figure 8.28, with the bonding sigma orbitals being populated through electron donation from the six ligand lone pairs. The eight metal *d* electrons fill the nonbonding t_{2g} orbitals, with one electron occupying each of the antibonding e_g^* orbitals. The electron spins are paired in the t_{2g} orbitals but are parallel to one another in the e_g^* orbitals, making the complex paramagnetic. There are no low energy π or π^* orbitals available on the ligands for π bonding. The center portion of the MO diagram is identical to that predicted by crystal field theory, as shown in Figure 8.17, and the magnetic properties are predicted to be the same by both methods. VB theory fails to describe bonding in this complex because it would need eight hybrid orbitals to accommodate the eight *d* electrons to form two-center two-electron bonds with the ligands. MO theory correctly accounts for the optical and magnetic properties of the complex.

Rare Earths 稀土 (RE)

1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	**															

Lanthanides 镧系元素

*	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
**	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Actinides 锕系元素

Rare earth = Sc + Y + Lanthanides

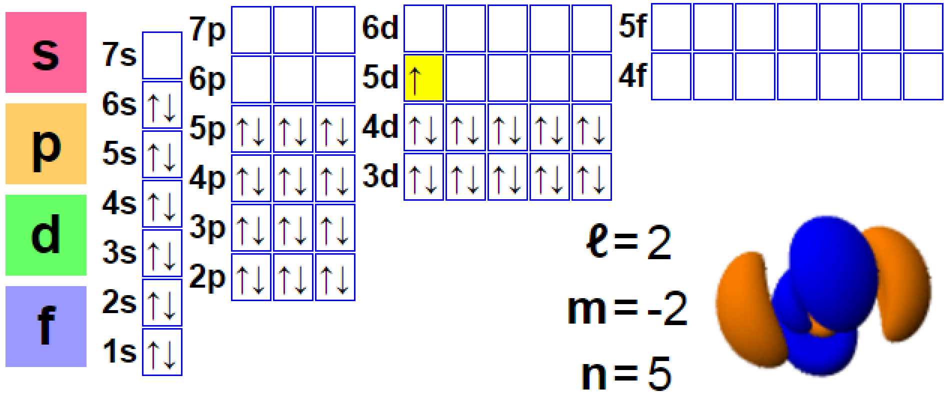
Electronic Configurations (1)

镧

57
La
Lanthanum
138.90547

2
8
18
18
9
2

2 **3**

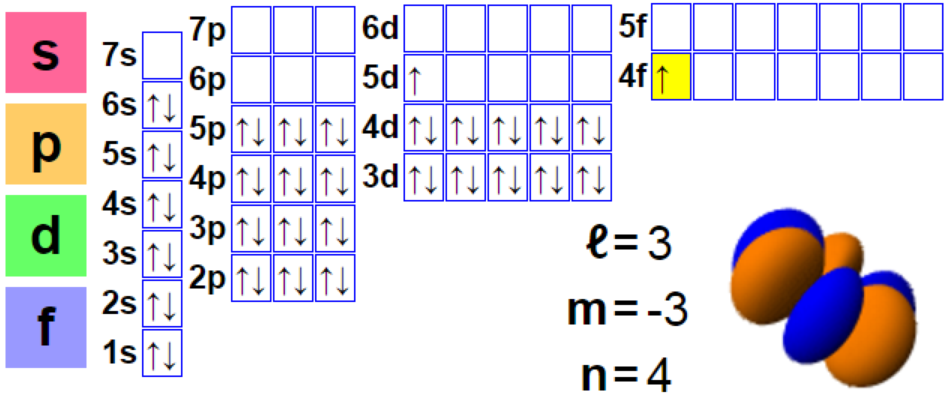


铈

58
Ce
Cerium
140.116

2
8
18
19
9
2

2 **3** **4**

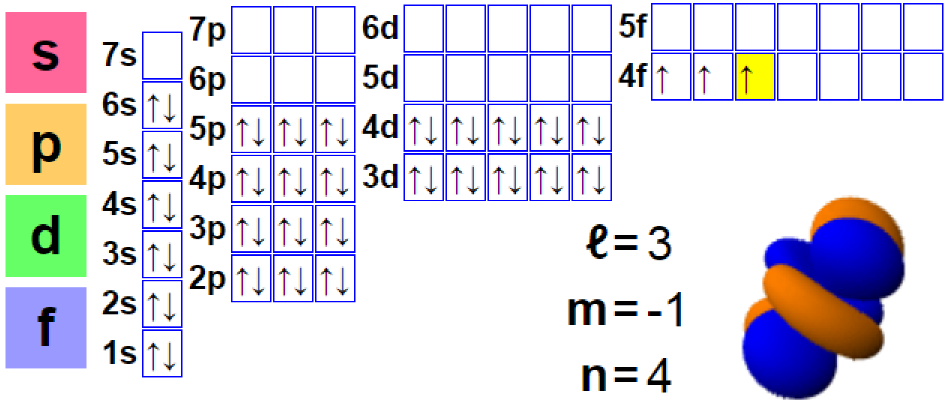


镨

59
Pr
Praseodymium
140.90765

2
8
18
21
8
2

2 **3** 4

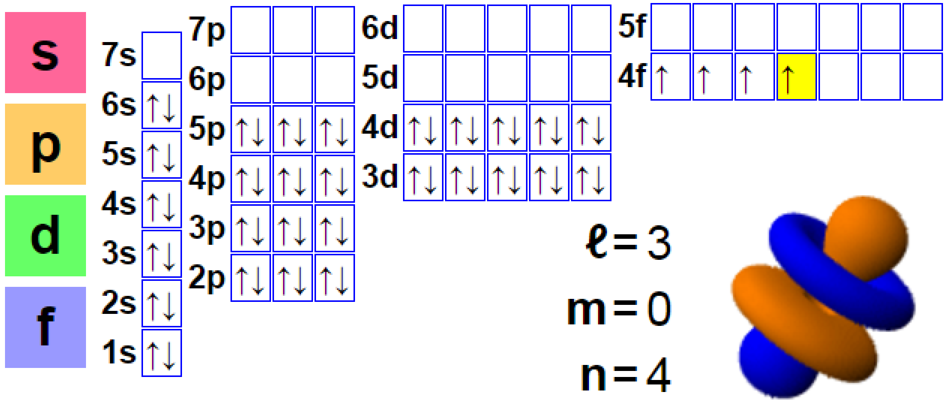


Electronic Configurations (2)

钕

60
Nd
Neodymium
144.242
2 8 18 22 8 2

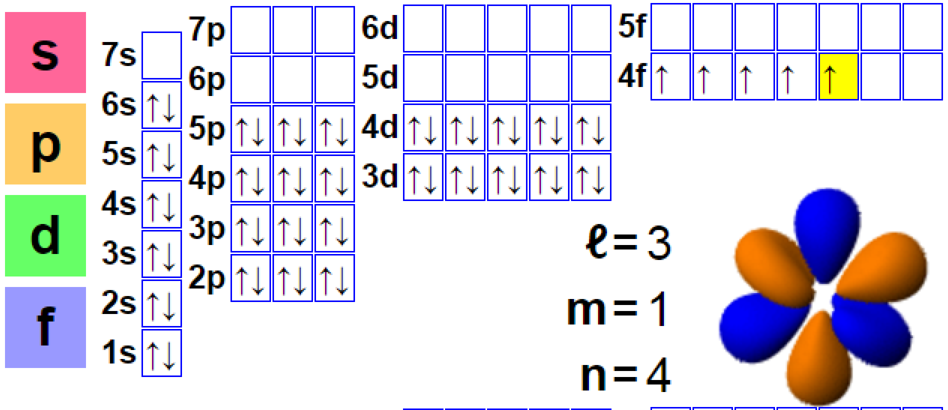
2 3



钷

61
Pm
Promethium
(145)
2 8 18 23 8 2

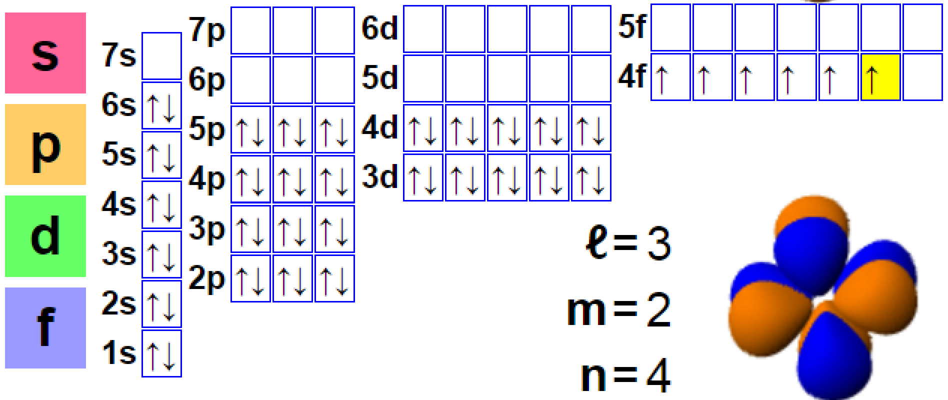
3



钐

62
Sm
Samarium
150.36
2 8 18 24 8 2

2 3



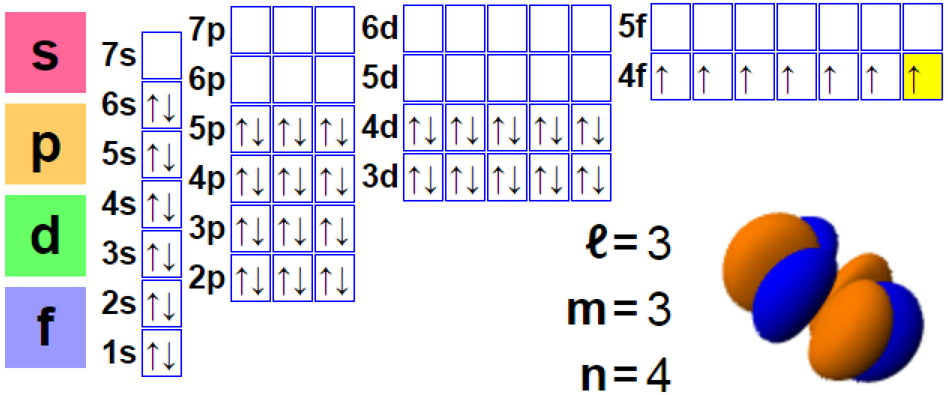
Electronic Configurations (3)

铕

63 Eu Europium 151.964

2 8 18 25 8 2

2 3

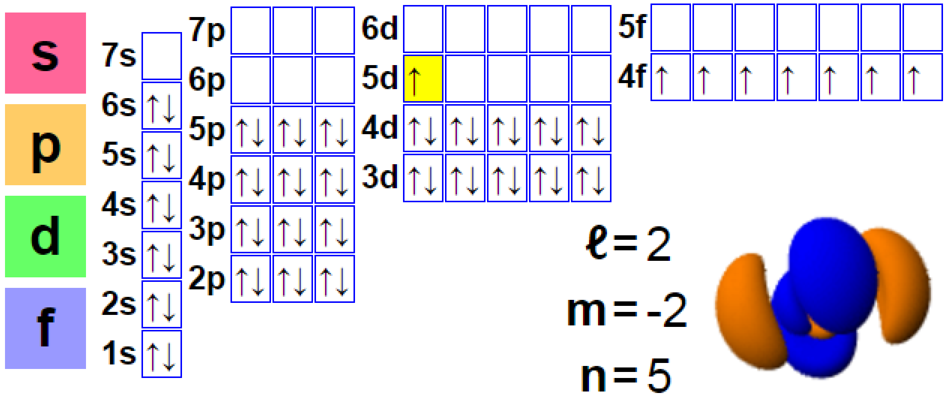


钆

64 Gd Gadolinium 157.25

2 8 18 25 9 2

1 2 3

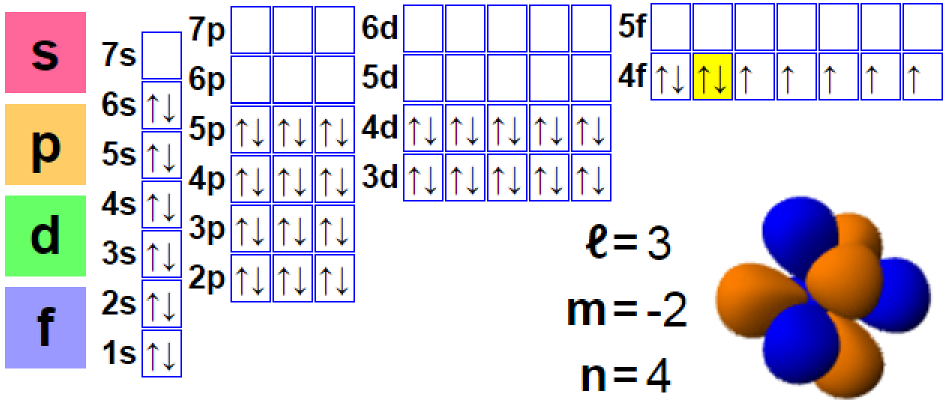


铽

65 Tb Terbium 158.92535

2 8 18 27 8 2

1 3 4



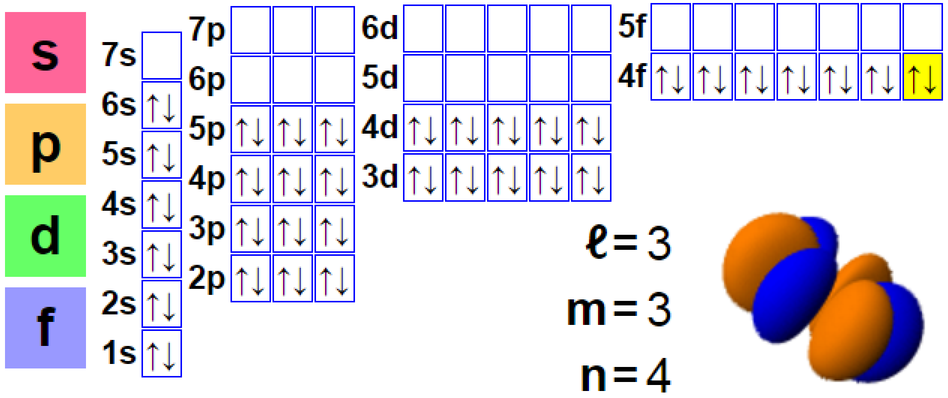
Electronic Configurations (4)

镱

70
Yb
Ytterbium
173.054

2
8
18
32
8
2

2 **3**

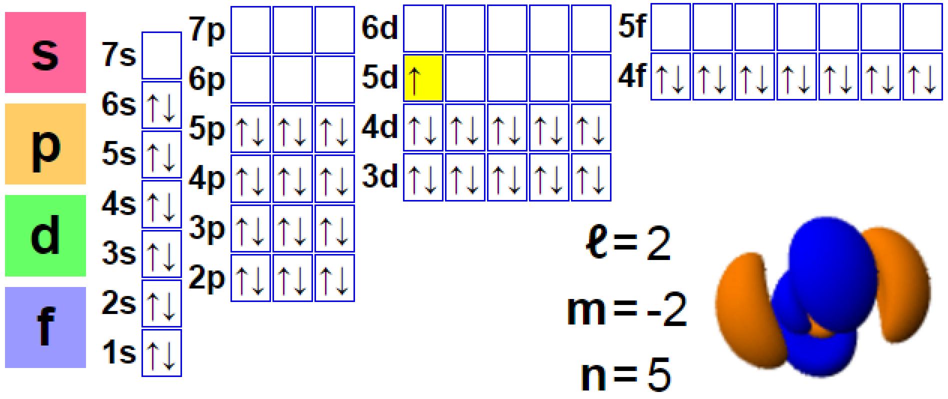


镥

71
Lu
Lutetium
174.9668

2
8
18
32
9
2

3



Oxidation States

Rare earth	La	Ce	Pr	Nd	Pm	Sm	Eu
$6s$+$5d$+$4f$ electrons	3	4	5	6	7	8	9
Oxidation states	+3	+3 +4	+3 +4	+3	+3	+2 +3	+2 +3
$4f$ electrons in RE^{n+}	0	1 0	2 1	3	4	6 5	7 6

Gd	Td	Dy	Ho	Er	Tm	Yb	Lu
10	11	12	13	14	15	16	17
+3	+3 +4	+3	+3	+3	+3	+2 +3	+3
7	8 7	9	10	11	12	14 13	14

Chemical Properties

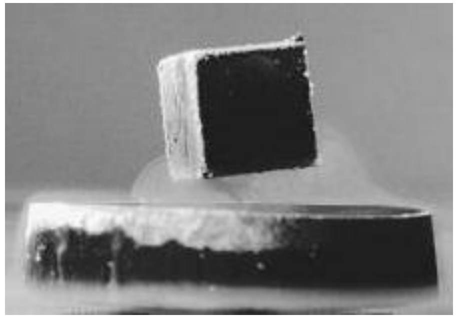
Rare earth	La	Ce	Pr	Nd	Pm	Sm	Eu
Oxidation states	+3	+3 +4	+3 +4	+3	+3	+2 +3	+2 +3
$r(RE^{3+})$ / pm	117.2	115	113	112.3	111	109.8	108.7
Category	Light rare earths				Radio-active	Middle REs	

Gd	Td	Dy	Ho	Er	Tm	Yb	Lu
+3	+3 +4	+3	+3	+3	+3	+2 +3	+3
107.8	106.3	105.2	104.1	103	102	100.8	100.1
Middle rare earths			Heavy rare earths				

Applications: Chemical

Forms very stable oxides and fluorides

- High m.p. and hardness
- Doping in functional ceramics and crystals



$YBa_2Cu_3O_{7-x}$

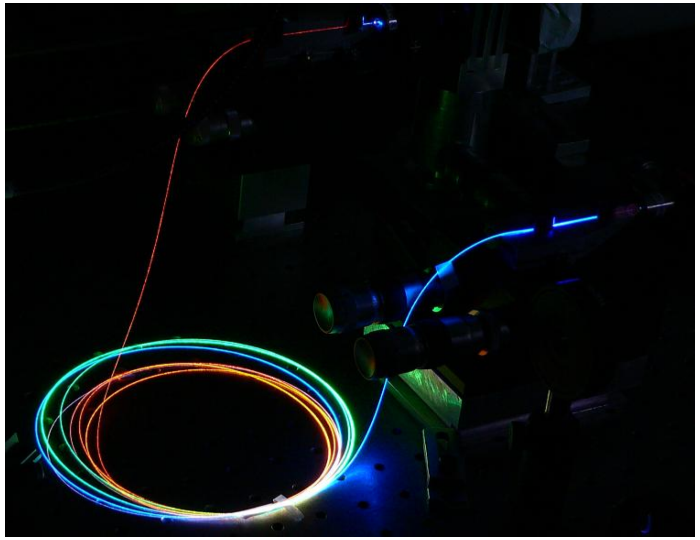
CeO_2
M.p. ~2400 °C

Applications: Functional Materials

The unpaired *f* electrons are unaffected by chemistry

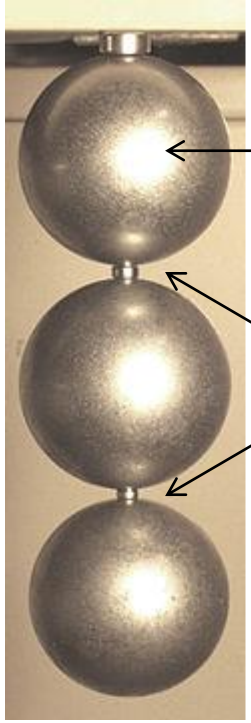
- Unlike crystal-field splitting for *d* electrons
- Optical and magnetic properties under extreme conditions

Nd:YAG laser crystals



Er-doped optical fibers

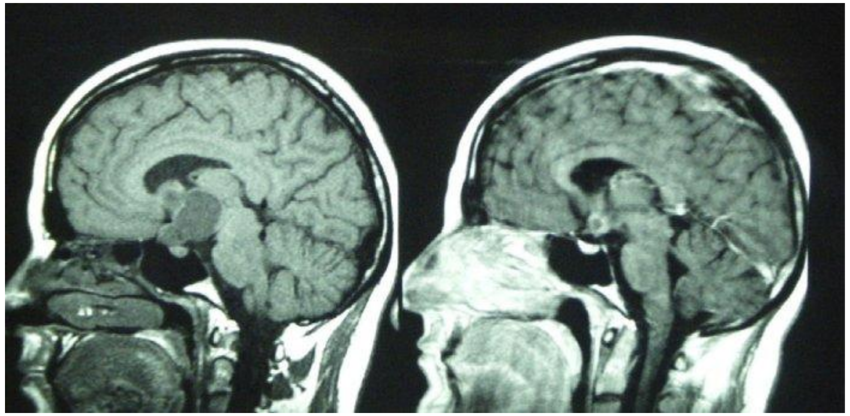
Applications: Functional Materials



Steel balls

NdFeB magnets

Electrical motors (Dy)



NMR imaging (Gd)

Homework: read “principle of modern chemistry”, chapter 8